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(54) **ELECTROCHEMICAL CELLS WITH MULTIPLE SEPARATORS, AND METHODS OF PRODUCING THE SAME**

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(57) **ABSTRACT**

Embodiments described herein relate to electrochemical cells with multiple separators, and methods of producing the same. A method of producing an electrochemical cell can include disposing an anode material onto an anode current collector, disposing a first separator on the anode material, disposing a cathode material onto a cathode current collector, disposing a second separator onto the cathode material, and disposing the first separator on the second separator to form the electrochemical cell. The anode material and/or the cathode material can be a semi-solid electrode material including an active material, a conductive material, and a volume of liquid electrolyte. In some embodiments, less than about 10% by volume of the liquid electrolyte evaporates during the forming of the electrochemical cell. In some embodiments, the method can further include wetting the first separator and/or the second separator with an electrolyte solution prior to coupling the first separator to the second separator.

24 Claims, 13 Drawing Sheets

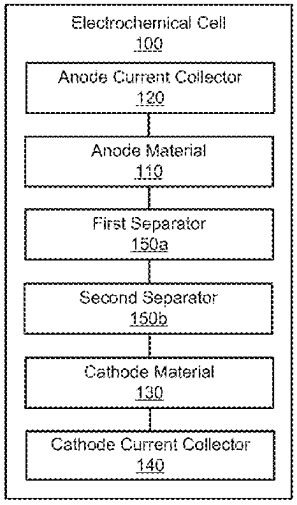

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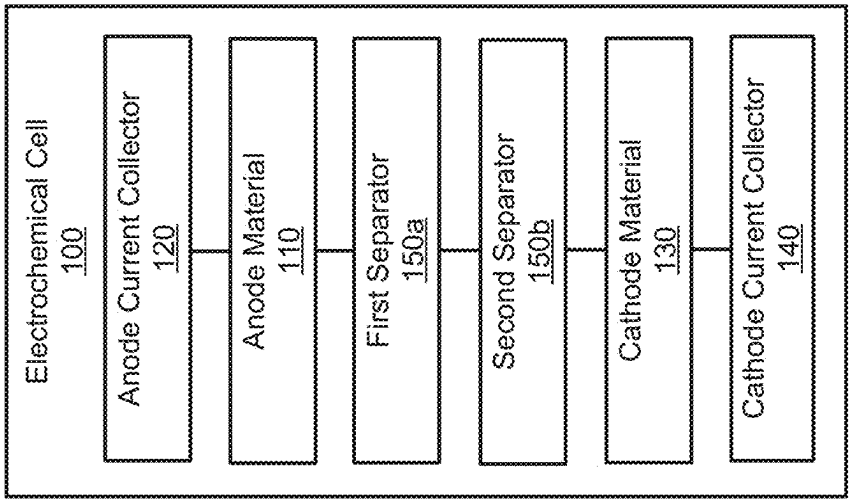
FIG. 1


Electrochemical Cell
100
Anode Current Collector
120
Anode Material
110
First Separator
150a
Second Separator
150b
Cathode Material
130
Cathode Current Collector
140

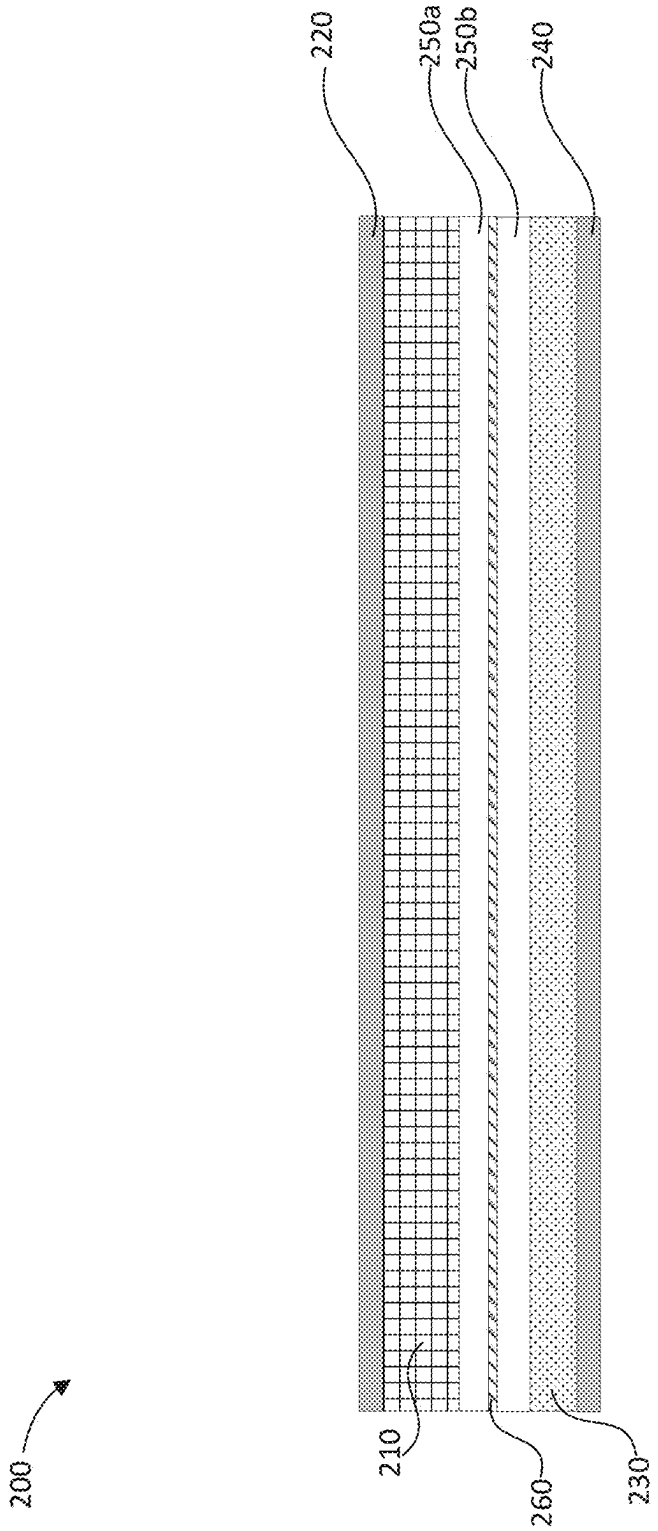


FIG. 2

FIG. 3

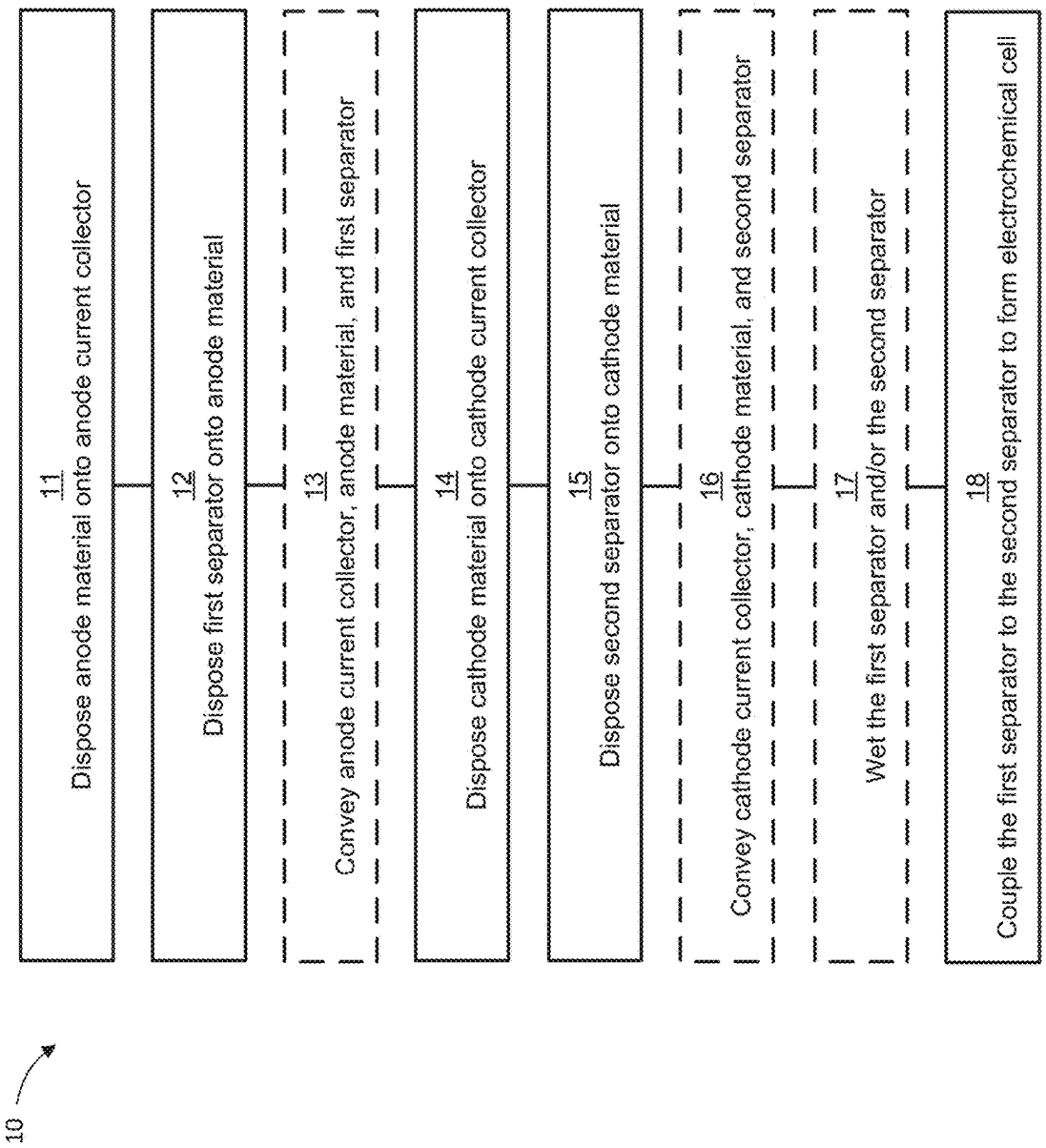

10
11
Dispose anode material onto anode current collector
12
Dispose first separator onto anode material
13
Convey anode current collector, anode material, and first separator
14
Dispose cathode material onto cathode current collector
15
Dispose second separator onto cathode material
16
Convey cathode current collector, cathode material, and second separator
17
Wet the first separator and/or the second separator
18
Couple the first separator to the second separator to form electrochemical cell

FIG. 4A

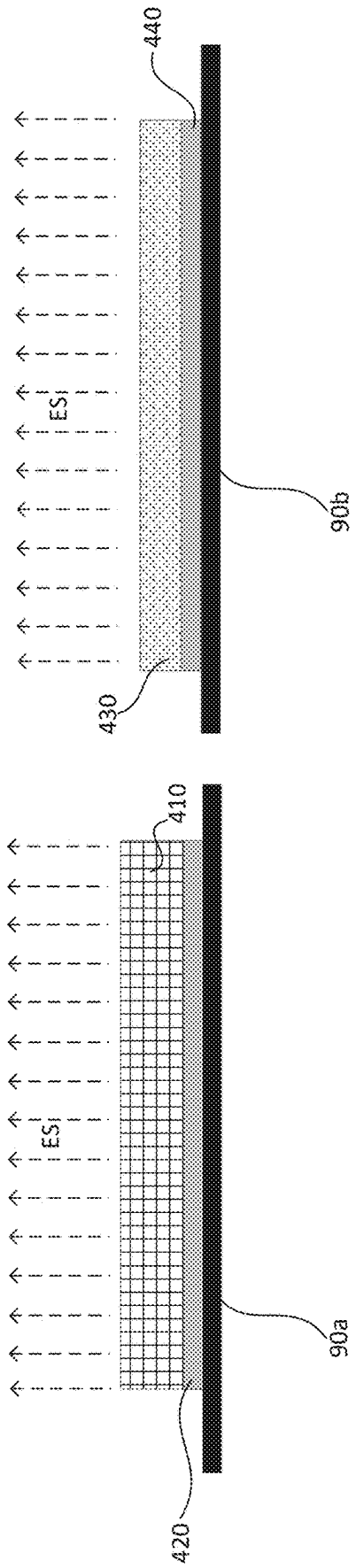


FIG. 4B

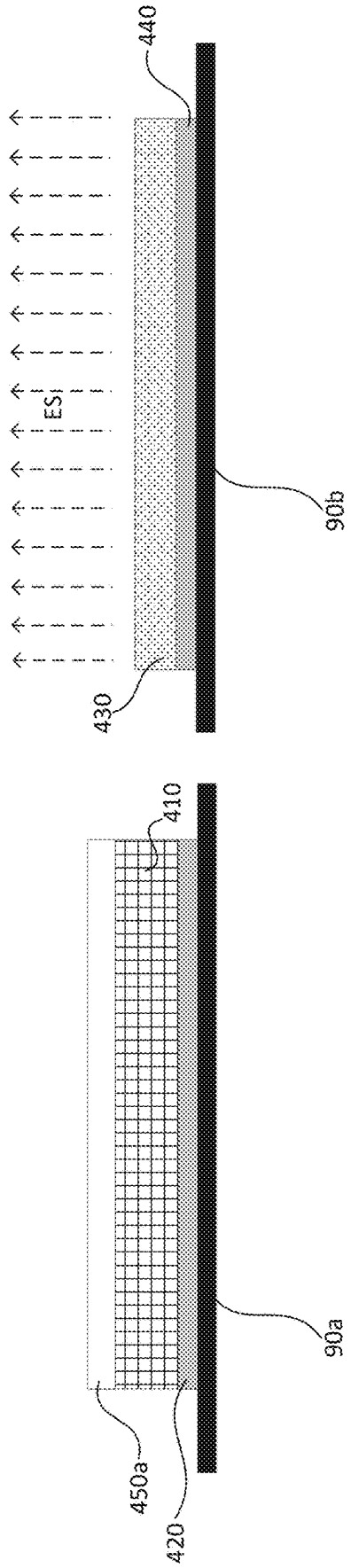


FIG. 4C

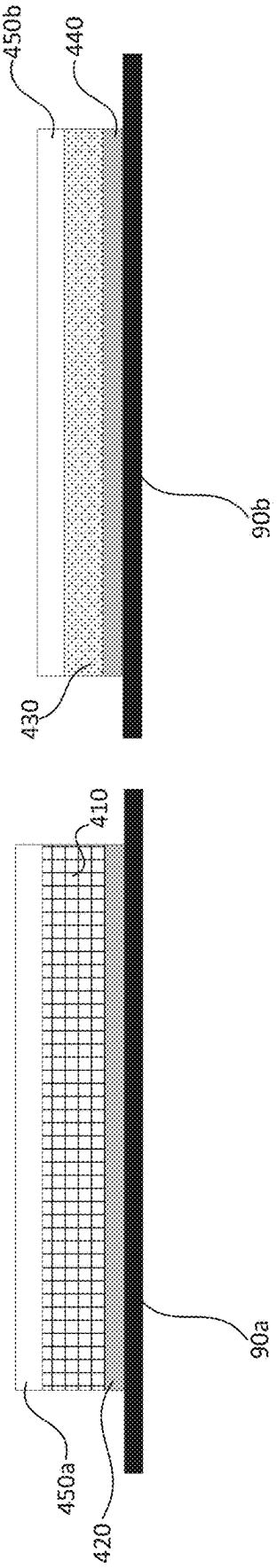


FIG. 5

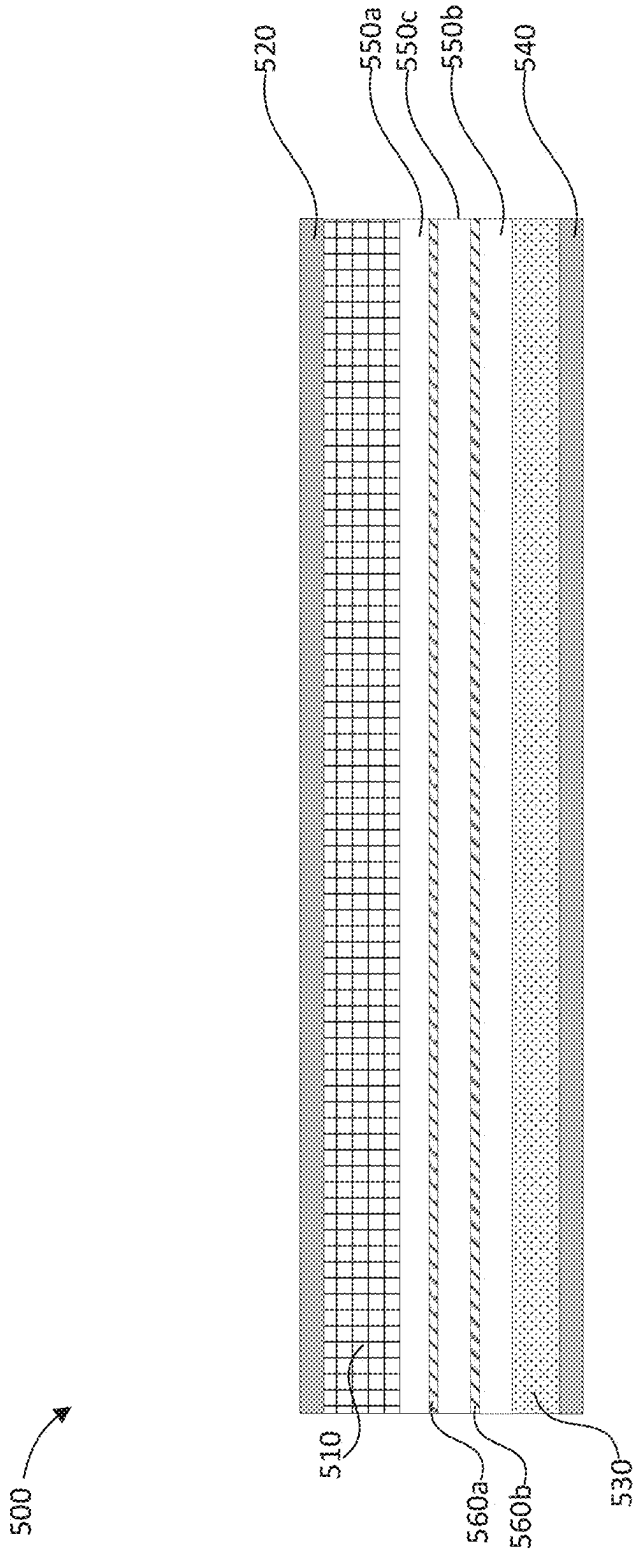
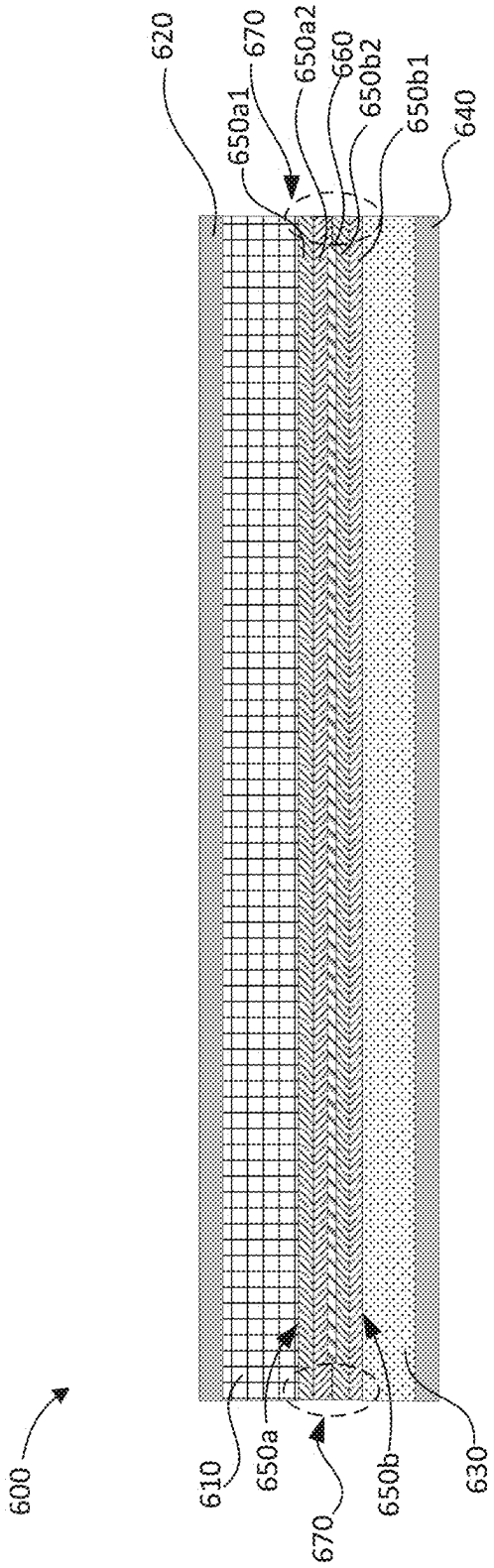


FIG. 6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650a
670
650b
630
620
650a1
670
650a2
660
650b2
650b1
640

FIG. 7

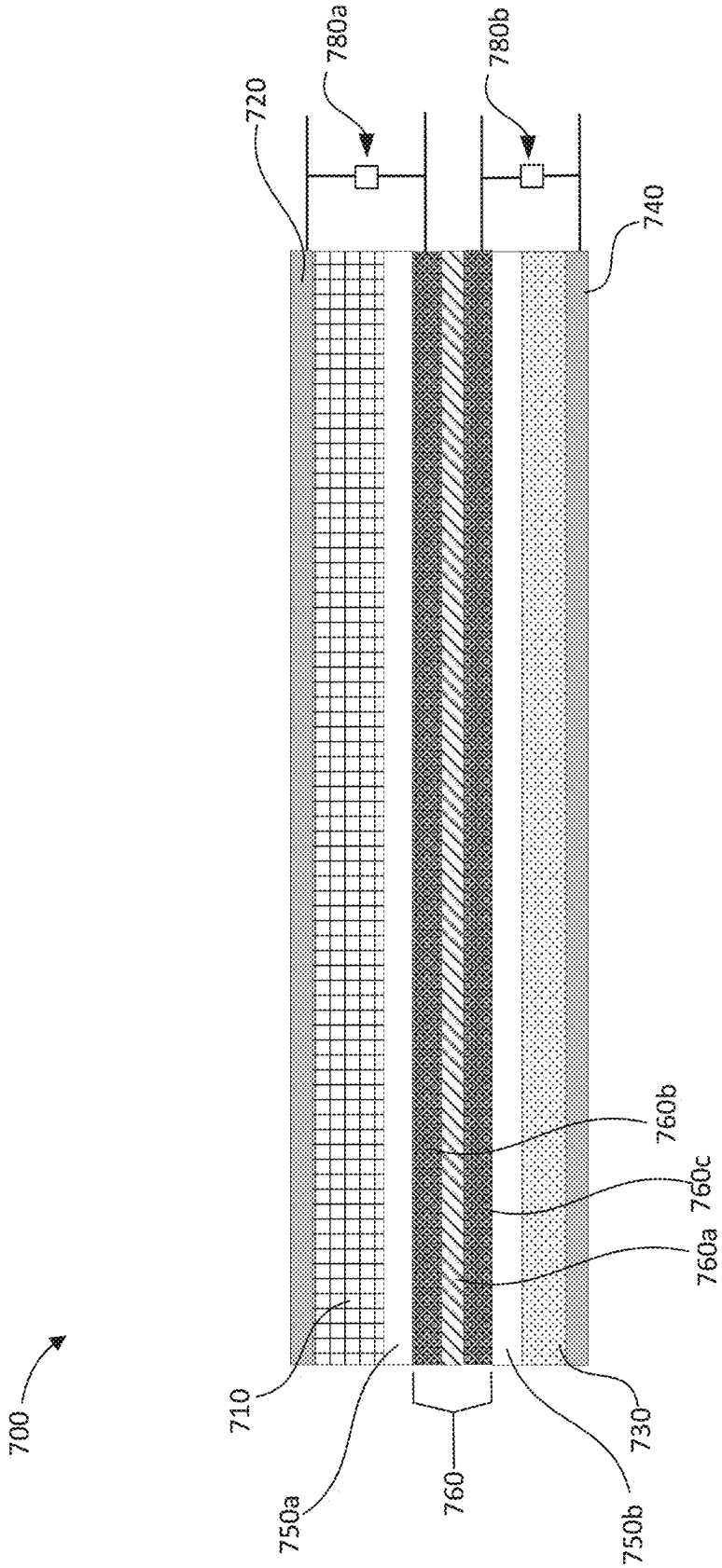


FIG. 8A

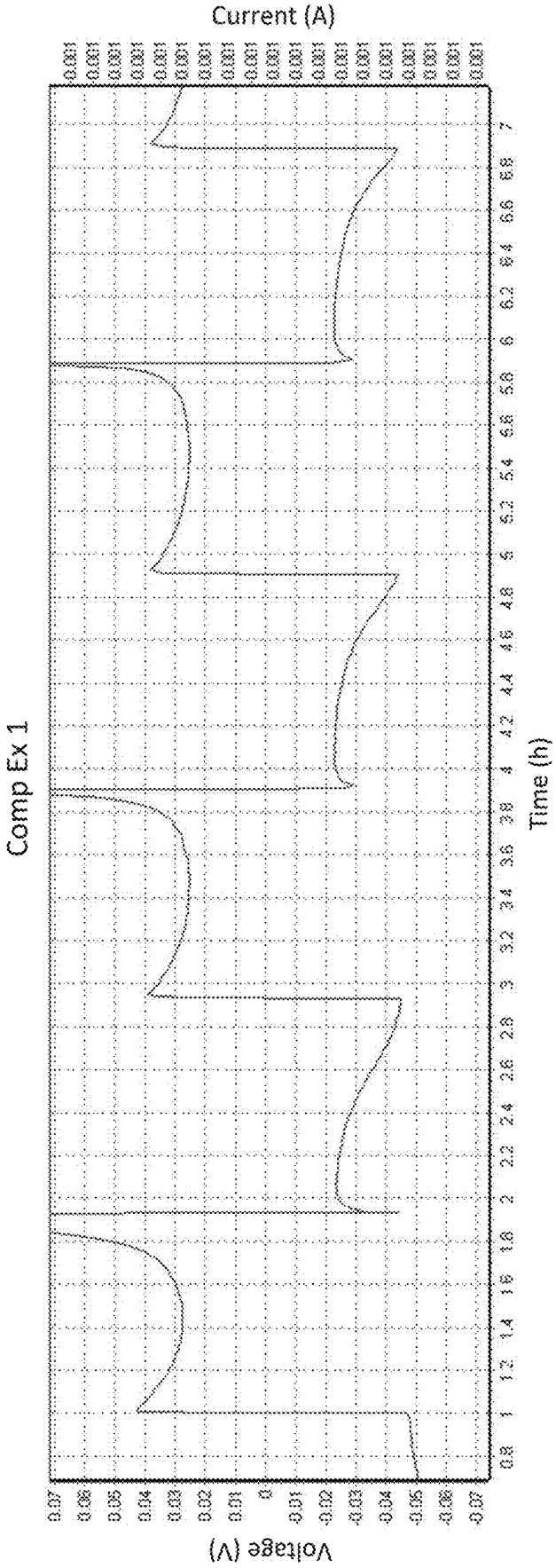


FIG. 8B

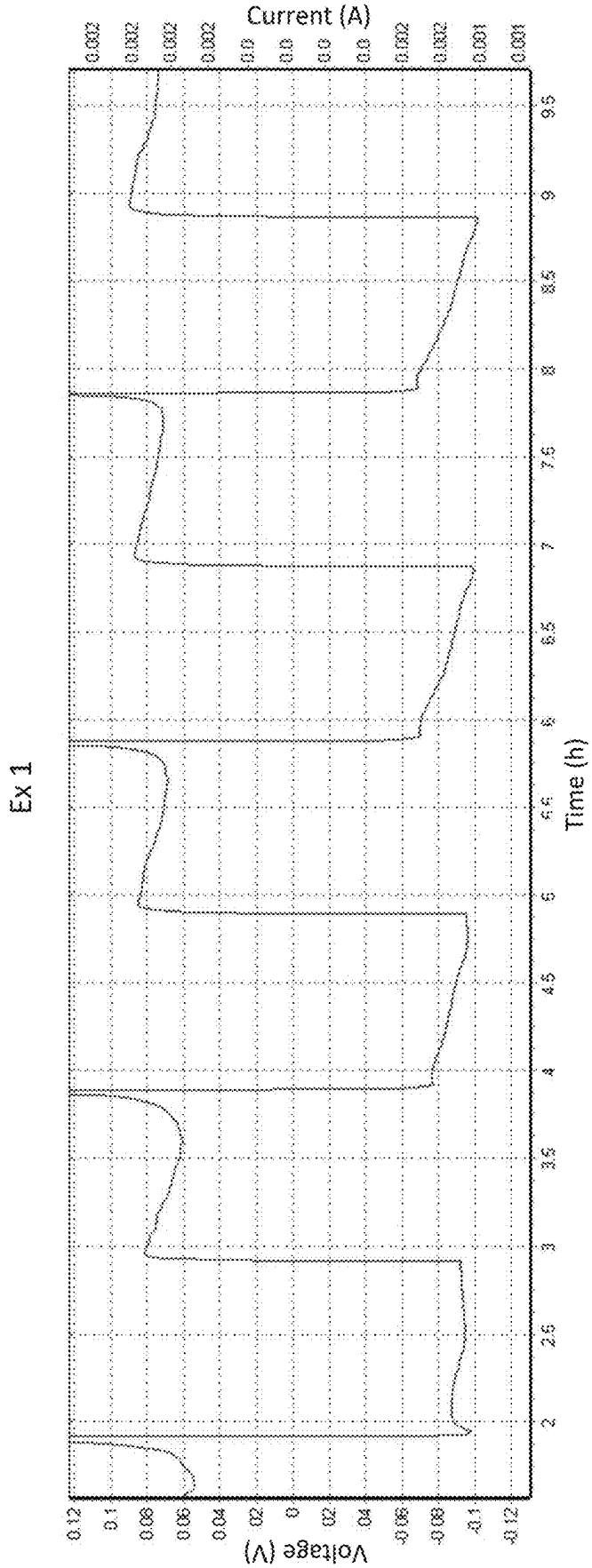

Ex 1
Current (A)
0.002
0.002
0.002
0.002
0.0
0.0
0.0
0.0
0.002
0.002
0.001
0.001
Voltage (V)
0.12
0.1
0.08
0.06
0.04
0.02
0
0.02
0.04
-0.06
-0.08
-0.1
-0.12
Time (h)
2
2.5
3
3.5
4
4.5
5
5.5
6
6.5
7
7.5
8
8.5
9
9.5

FIG. 9A

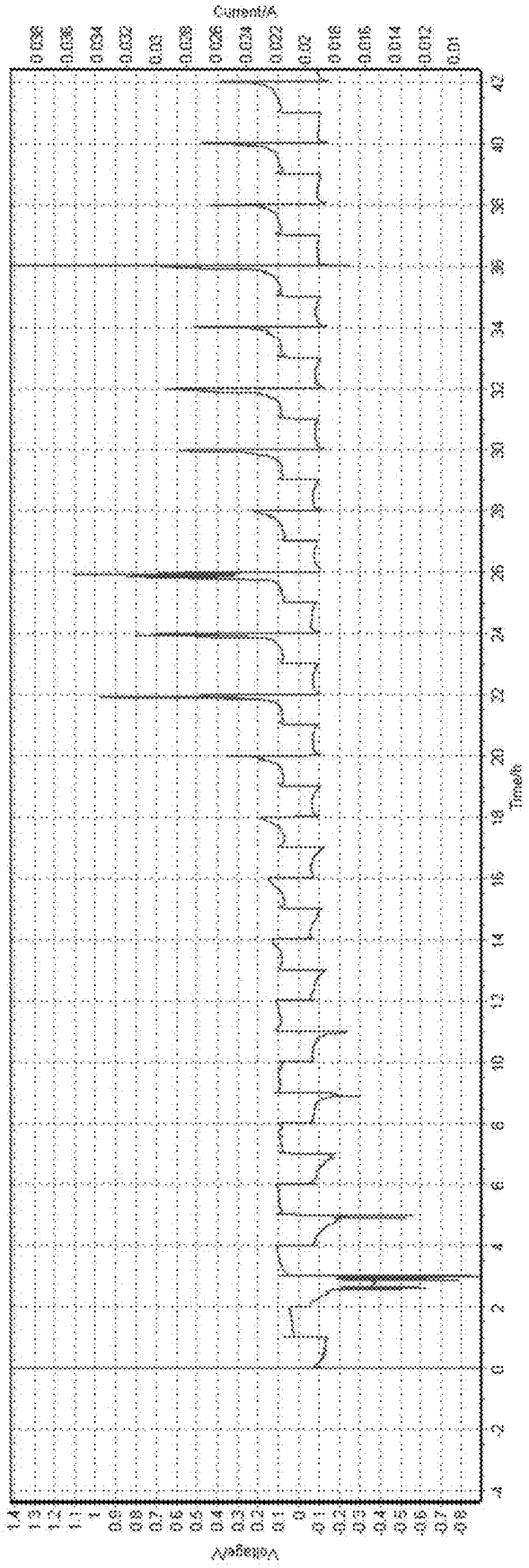
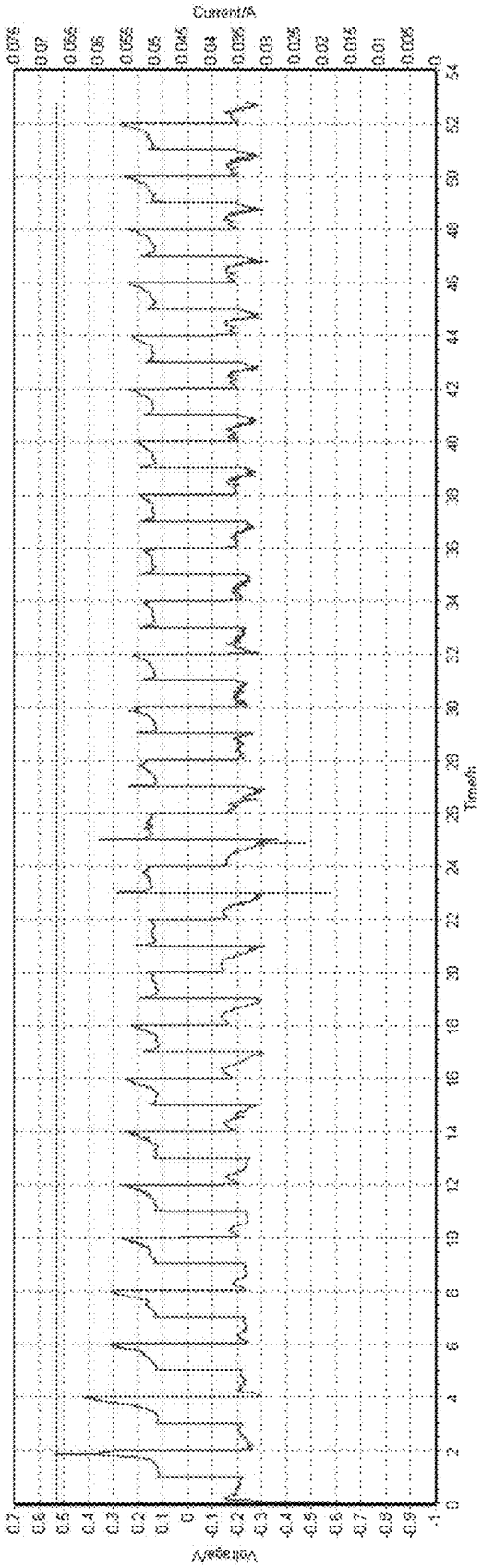


FIG. 9B



ELECTROCHEMICAL CELLS WITH MULTIPLE SEPARATORS, AND METHODS OF PRODUCING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 17/733,658, filed Apr. 29, 2022, and entitled "Electrochemical Cells with Multiple Separators, and Methods of Producing the Same," which claims priority to and benefit of U.S. Provisional Application No. 63/181,721, filed Apr. 29, 2021, and entitled "Electrochemical Cells with Multiple Separators and Methods of Producing the Same," the entire disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

Embodiments described herein relate to electrochemical cells with multiple separators, and methods of producing the same.

BACKGROUND

Electrolyte is added to electrodes during production of electrochemical cells. Electrolyte is often in the form of an electrolyte solvent with an electrolyte salt dissolved therein. Conventional electrochemical cell production processes include forming solid electrodes, placing them in a container and adding the electrolyte to the container. However, formation of semi-solid electrodes can include adding an electrolyte solution to an active material and a conductive material to form a slurry. During the production process, the slurry can be moved from one location to another, and electrolyte solvent can evaporate from the slurry. This solvent can be costly to replace. Preventing solvent evaporation rather than replacing evaporated solvent can significantly reduce costs associated with production of electrochemical cells.

SUMMARY

Embodiments described herein relate to electrochemical cells with multiple separators, and methods of producing the same. A method of producing an electrochemical cell can include disposing an anode material onto an anode current collector, disposing a first separator on the anode material, disposing a cathode material onto a cathode current collector, disposing a second separator onto the cathode material, and disposing the first separator on the second separator to form the electrochemical cell. The anode material and/or the cathode material can be a semi-solid electrode material including an active material, a conductive material, and a volume of liquid electrolyte. In some embodiments, less than about 10% by volume of the liquid electrolyte evaporates during the forming of the electrochemical cell. In some embodiments, the method can further include wetting the first separator and/or the second separator with an electrolyte solution prior to coupling the first separator to the second separator. In some embodiments, the wetting is via spraying. In some embodiments, less than about 10% by volume of the electrolyte solution evaporates during the forming of the electrochemical cell. In some embodiments, less than about 10% of a total volume of a combination of the electrolyte solution and the liquid electrolyte can evaporate during the forming of the electrochemical cell. In some embodiments, the first separator and/or the second separator can be composed of a material with a porosity of less than about 1%. In some embodiments, the cathode current collector, the cathode material, and the second separator can collectively form a cathode, and the method further comprises conveying the cathode along a cathode conveyor. In some embodiments, the anode current collector, the anode material, and the first separator can collectively form an anode, and the method further comprises conveying the anode along an anode conveyor. In some embodiments, the anode conveyor can be the same conveyor as the cathode conveyor. In some embodiments, the anode conveyor can be a different conveyor from the cathode conveyor.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a block diagram an electrochemical cell with multiple separators, according to an embodiment.

FIG. **2** is an illustration of an electrochemical cell with multiple separators, according to an embodiment.

FIG. **3** is a block diagram of a method of manufacturing an electrochemical cell with multiple separators, according to an embodiment.

FIGS. **4**A-**4**C show methods of conveyance of electrodes, according to various embodiments.

FIG. **5** is an illustration of an electrochemical cell with multiple separators, according to an embodiment.

FIG. **6** is an illustration of an electrochemical cell with multiple separators, according to an embodiment.

FIG. **7** is an illustration of an electrochemical cell with multiple separators, according to an embodiment.

FIGS. **8**A-**8**B show a comparison between a control electrochemical cell with a single separator and an electrochemical cell with two separators and a layer of hard carbon disposed therebetween.

FIGS. **9**A-**9**B show a comparison between a control electrochemical cell with a single separator and an electrochemical cell with two separators and a layer of hard carbon disposed therebetween.

DETAILED DESCRIPTION

Embodiments described herein relate to multi-separator electrochemical cells and systems, and methods of manufacturing the same. Separators in electrochemical cells physically isolate an anode from a cathode so as to prevent short circuits and maintain a voltage difference between the anode and the cathode. Pores in separators allow passage of electroactive species therethrough. Separators can have an additional benefit of shielding evaporation of electrolyte solution during production.

In some embodiments, electrodes described herein can be semi-solid electrodes. In comparison to conventional electrodes, semi-solid electrodes can be made (i) thicker (e.g., greater than about 250 μm-up to about 2,000 μm or even greater) due to the reduced tortuosity and higher electronic conductivity of semi-solid electrodes, (ii) with higher loadings of active materials, (iii) with a simplified manufacturing process utilizing less equipment, and (iv) can be operated between a wide range of C-rates while maintaining a substantial portion of their theoretical charge capacity. These relatively thick semi-solid electrodes decrease the volume, mass and cost contributions of inactive components with respect to active components, thereby enhancing the commercial appeal of batteries made with the semi-solid electrodes. In some embodiments, the semi-solid electrodes described herein, are binderless and/or do not use binders

that are used in conventional battery manufacturing. Instead, the volume of the electrode normally occupied by binders in conventional electrodes, is now occupied, by: 1) electrolyte, which has the effect of decreasing tortuosity and increasing the total salt available for ion diffusion, thereby countering the salt depletion effects typical of thick conventional electrodes when used at high rate, 2) active material, which has the effect of increasing the charge capacity of the battery, or 3) conductive additive, which has the effect of increasing the electronic conductivity of the electrode, thereby countering the high internal impedance of thick conventional electrodes. The reduced tortuosity and a higher electronic conductivity of the semi-solid electrodes described herein, results in superior rate capability and charge capacity of electrochemical cells formed from the semi-solid electrodes.

Since the semi-solid electrodes described herein can be made substantially thicker than conventional electrodes, the ratio of active materials (i.e., the semi-solid cathode and/or anode) to inactive materials (i.e., the current collector and separator) can be much higher in a battery formed from electrochemical cell stacks that include semi-solid electrodes relative to a similar battery formed form electrochemical cell stacks that include conventional electrodes. This substantially increases the overall charge capacity and energy density of a battery that includes the semi-solid electrodes described herein. The use of semi-solid, binderless electrodes can also be beneficial in the incorporation of an overcharge protection mechanism, as generated gas can migrate to the electrode/current collector interface without binder particles inhibiting the movement of the gas within the electrode.

In some embodiments, the electrode materials described herein can be a flowable semi-solid or condensed liquid composition. A flowable semi-solid electrode can include a suspension of an electrochemically active material (anodic or cathodic particles or particulates), and optionally an electronically conductive material (e.g., carbon) in a non-aqueous liquid electrolyte. Said another way, the active electrode particles and conductive particles are co-suspended in a liquid electrolyte to produce a semi-solid electrode. Examples of electrochemical cells that include a semi-solid and/or binderless electrode material are described in U.S. Pat. No. 8,993,159 entitled, "Semi-solid Electrodes Having High Rate Capability," filed Apr. 29, 2013 ("the '159 patent"), the disclosure of which is incorporated herein by reference in its entirety.

In some embodiments, the electrode materials described herein can be a flowable semi-solid or condensed liquid composition. In some embodiments, a flowable semi-solid electrode can include a suspension of an electrochemically active material (anodic or cathodic particles or particulates), and optionally an electronically conductive material (e.g., carbon) in a non-aqueous liquid electrolyte. In some embodiments, the active electrode particles and conductive particles can be co-suspended in an electrolyte to produce a semi-solid electrode. In some embodiments, electrode materials described herein can include conventional electrode materials (e.g., including lithium metal).

Semi-solid electrodes have a liquid electrolyte integrated therein during a longer portion of the manufacturing process than conventional electrodes, which add electrolyte solution after the electrodes are fully formed. In other words, liquid electrolyte is added to conductive materials and/or active materials to form a semi-solid electrode material. While the semi-solid electrode material is undergoing further processing, liquid electrolyte solvent can evaporate from the semi-solid electrode material. This evaporation can raise the molarity of electrolyte salt in the electrolyte solution, potentially causing salt buildup. Built-up salt can prevent passage of electroactive species through the semi-solid electrode material. In other words, movement of electroactive species through pores of the semi-solid electrode material can be more difficult when salt ions build up and block flow paths. Additionally, evaporation of electrolyte solution can make the semi-solid electrode material less flowable and/or less malleable. Liquid flow paths within the semi-solid electrode material can dry out, increasing tortuosity of the movements of electroactive species.

While adding electrolyte solvent during production can address some of these problems, make-up electrolyte solvent can add significant cost to the production process. Coupling separators to the anode and/or the cathode during production of the electrochemical cell can aid in reducing evaporation of electrolyte solvent during production. In some embodiments, separators described herein can have geometries and general properties the same or substantially similar to those described in PCT Application US2020/058564 entitled "Electrochemical Cells with Separator Seals, and Methods of Manufacturing the Same," filed Nov. 2, 2020 ("the '564 application"), the disclosure of which is hereby incorporated by reference in its entirety.

FIG. **1** is a block diagram of an electrochemical cell **100** with multiple separators, according to an embodiment. As shown, the electrochemical cell **100** includes an anode material **110** disposed on an anode current collector **120** and a cathode material **130** disposed on a cathode current collector **140**, with a first separator **150***a* and a second separator **150***b* (collectively referred to as "separators **150**") disposed therebetween. In some embodiments, the anode material **110** and/or the cathode material **130** can include a semi-solid electrode material. In some embodiments, the anode material **110** and/or the cathode material **130** can include any of the properties of the semi-solid electrodes described in the '159 patent.

In some embodiments, the anode material **110** and/or the cathode material **130** can include at least about 0.1%, at least about 0.2%, at least about 0.3%, at least about 0.4%, at least about 0.5%, at least about 0.6%, at least about 0.7%, at least about 0.8%, at least about 0.9%, at least about 1%, at least about 2%, at least about 3%, at least about 4%, at least about 5%, at least about 6%, at least about 7%, at least about 8%, at least about 9%, at least about 10%, at least about 11%, at least about 12%, at least about 13%, at least about 14%, at least about 15%, at least about 16%, at least about 17%, at least about 18%, at least about 19%, at least about 20%, at least about 21%, at least about 22%, at least about 23%, or at least about 24% by volume of liquid electrolyte solution. In some embodiments, the anode material **110** and/or the cathode material **130** can include no more than about 25%, no more than about 24%, no more than about 23%, no more than about 22%, no more than about 21%, no more than about 20%, no more than about 19%, no more than about 18%, no more than about 17%, no more than about 16%, no more than about 15%, no more than about 14%, no more than about 13%, no more than about 12%, no more than about 11%, no more than about 10%, no more than about 9%, no more than about 8%, no more than about 7%, no more than about 6%, no more than about 5%, no more than about 4%, no more than about 3%, no more than about 2%, no more than about 1%, no more than about 0.9%, no more than about 0.8%, no more than about 0.7%, no more than about 0.6%, no more than about 0.5%, no more than about 0.4%, no more than about 0.3%, or no more than about 0.2% by volume of liquid electrolyte solution.

Combinations of the above-referenced volumetric percentages of liquid electrolyte solution in the anode material **110** and/or the cathode material **130** are also possible (e.g., at least about 0.1% and no more than about 25% or at least about 5% and no more than about 10%), inclusive of all values and ranges therebetween. In some embodiments, the anode material **110** and/or the cathode material **130** can include about 0.1%, about 0.2%, about 0.3%, about 0.4%, about 0.5%, about 0.6%, about 0.7%, about 0.8%, about 0.9%, about 1%, about 2%, about 3%, about 4%, about 5%, about 6%, about 7%, about 8%, about 9%, about 10%, about 11%, about 12%, about 13%, about 14%, about 15%, about 16%, about 17%, about 18%, about 19%, about 20%, about 21%, about 22%, about 23%, about 24%, or about 25% by volume of liquid electrolyte solution.

In some embodiments, the anode current collector **120** and/or the cathode current collector **140** can be composed of copper, aluminum, titanium, or other metals that do not form alloys or intermetallic compounds with lithium, carbon, and/or coatings comprising such materials disposed on another conductor. In some embodiments, the anode current collector **120** and/or the cathode current collector **140** can have a thickness of at least about 1 μm, at least about 5 μm, at least about 10 μm, at least about 15 μm, at least about 20 μm, at least about 25 μm, at least about 30 μm, at least about 35 μm, at least about 40 μm, or at least about 45 μm. In some embodiments, the anode current collector **120** and/or the cathode current collector **140** can have a thickness of no more than about 50 μm, no more than about 45 μm, no more than about 40 μm, no more than about 35 μm, no more than about 30 μm, no more than about 25 μm, no more than about 20 μm, no more than about 15 μm, no more than about 10 μm, or no more than about 5 μm. Combinations of the above-referenced thicknesses of the anode current collector **120** and/or the cathode current collector **140** are also possible (e.g., at least about 1 μm and no more than about 50 μm or at least about 10 μm and no more than about 30 μm), inclusive of all values and ranges therebetween. In some embodiments, the anode current collector **120** and/or the cathode current collector **140** can have a thickness of about 1 μm, about 5 μm, about 10 μm, about 15 μm, about 20 μm, about 25 μm, about 30 μm, about 35 μm, about 40 μm, about 45 μm, or about 50 μm.

In some embodiments, the anode material **110** can include a first electrolyte and the cathode material **130** can include a second electrolyte. In other words, and the anode material **110** can include an anolyte and the cathode material **130** can include a catholyte. In some embodiments, the electrochemical cell **100** can include an anolyte disposed on the anode side of the separators **150**. In some embodiments, the electrochemical cell **100** can include a catholyte disposed on the cathode side of the separators **150**. In some embodiments, the electrochemical cell **100** can include a selectively permeable membrane. In some embodiments, the selectively permeable membrane can be disposed between the first separator **150***a* and the second separator **150***b*. Electrochemical cells with anolytes, catholytes, and/or selectively permeable membranes are described in U.S. Pat. No. 10,734,672 ("the '672 patent"), filed Jan. 8, 2019, and titled, "Electrochemical Cells Including Selectively Permeable Membranes, Systems and Methods of Manufacturing the Same," the disclosure of which is hereby incorporated by reference in its entirety.

As shown, the first separator **150***a* is disposed on the anode material **110** while the second separator **150***b* is disposed on the cathode material **130**. In some embodiments, the separators **150** can be disposed on their respective electrodes during production of the electrochemical cell **100**. In some embodiments, the first separator **150***a* and/or the second separator **150***b* can be composed of polyethylene, polypropylene, high density polyethylene, polyethylene terephthalate, polystyrene, a thermosetting polymer, hard carbon, a thermosetting resin, a polyimide, a ceramic coated separator, an inorganic separator, cellulose, glass fiber, a polyethylenoxide (PEO) polymer in which a lithium salt is complexed to provide lithium conductivity, Nation™ membranes which are proton conductors, or any other suitable separator material, or combinations thereof. In some embodiments, the first separator **150***a* and/or the second separator **150***b* can be composed of any of the separator materials described in the '564 application. In some embodiments, the first separator **150***a* can be composed of the same material as the second separator **150***b*. In some embodiments, the first separator **150***a* can be composed of a different material from the second separator **150***b*. In some embodiments, the first separator **150***a* and/or the second separator **150***b* can be absent of any framing members described in the '564 application.

In some embodiments, the first separator **150***a* and/or the second separator **150***b* can have a porosity of at least about 5%, at least about 10%, at least about 15%, at least about 20%, at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, or at least about 85%. In some embodiments, the first separator **150***a* and/or the second separator **150***b* can have a porosity of no more than about 90%, no more than about 85%, no more than about 80%, no more than about 75%, no more than about 70%, no more than about 65%, no more than about 60%, no more than about 55%, no more than about 50%, no more than about 45%, no more than about 40%, no more than about 35%, no more than about 30%, no more than about 25%, no more than about 20%, no more than about 15%, or no more than about 10%.

Combinations of the above-referenced porosity percentages of the first separator **150***a* and/or the second separator **150***b* are also possible (e.g., at least about 5% and no more than about 90% or at least about 20% and no more than about 40%), inclusive of all values and ranges therebetween. In some embodiments, the first separator **150***a* and/or the second separator **150***b* can have a porosity of about 5%, about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or about 90%.

In some embodiments, the first separator **150***a* can have a different porosity from the second separator **150***b*. In some embodiments, the porosities of the first separator **150***a* and the second separator **150***b* can be selected based on the difference between the anolyte and the catholyte. For example, if the catholyte has a higher vapor pressure and faster evaporation properties than the anolyte, then the second separator **150***b* can have a lower porosity than the first separator **150***a*. The lower porosity of the second separator **150***b* can at least partially prevent the catholyte from evaporating during production.

In some embodiments, the first separator **150***a* can be composed of a different material from the second separator **150***b*. In some embodiments, the materials of the first separator **150***a* and the second separator **150***b* can be selected to facilitate wettability of the first separator **150***a* with the anolyte and the second separator **150***b* with the catholyte **150**. For example, an ethylene carbonate/propyl-

ene carbonate-based catholyte can wet a polyethylene separator better than a polyimide separator, based on the molecular properties of the materials. An ethylene carbonate/dimethyl carbonate-based anolyte can wet a polyimide separator better than a polyethylene separator. A full wetting of the first separator **150***a* and the second separator **150***b* can give way to better transport of electroactive species via the separators **150**. This transport can be facilitated particularly well when the first separator **150***a* physically contacts the second separator **150***b*.

In some embodiments, the first separator **150***a* and/or the second separator **150***b* can be absent of separator seals (e.g., separator seals described in the '564 application). As shown, the electrochemical cell **100** includes two separators **150**. In some embodiments, the electrochemical cell **100** can include 3, 4, 5, 6, 7, 8, 9, 10, or more than about 10 separators **150**. In some embodiments, a layer of liquid electrolyte (not shown) can be disposed between the first separator **150***a* and the second separator **150***b*. A layer of liquid electrolyte can promote better adhesion between the separators **150**.

In some embodiments, the first separator **150***a* and/or the second separator **150***b* can have a thickness of at least about 1 μm, at least about 2 μm, at least about 3 μm, at least about 4 μm, at least about 5 μm, at least about 6 μm, at least about 7 μm, at least about 8 μm, at least about 9 μm, at least about 10 μm, at least about 20 μm, at least about 30 μm, at least about 40 μm, at least about 50 μm, at least about 60 μm, at least about 70 μm, at least about 80 μm, at least about 90 μm, at least about 100 μm, at least about 110 μm, at least about 120 μm, at least about 130 μm, at least about 140 μm, at least about 150 μm, at least about 160 μm, at least about 170 μm, at least about 180 μm, or at least about 190 μm. In some embodiments, the first separator **150***a* and/or the second separator **150***b* can have a thickness of no more than about 200 μm, no more than about 190 μm, no more than about 180 μm, no more than about 170 μm, no more than about 160 μm, no more than about 150 μm, no more than about 140 μm, no more than about 130 μm, no more than about 120 μm, no more than about 110 μm, no more than about 100 μm, no more than about 90 μm, no more than about 80 μm, no more than about 70 μm, no more than about 60 μm, no more than about 50 μm, no more than about 40 μm, no more than about 30 μm, no more than about 20 μm, no more than about 10 μm, no more than about 9 μm, no more than about 8 μm, no more than about 7 μm, no more than about 6 μm, no more than about 5 μm, no more than about 4 μm, no more than about 3 μm, or no more than about 2 μm. Combinations of the above-referenced thicknesses of the first separator **150***a* and/or the second separator **150***b* are also possible (e.g., at least about 1 μm and no more than about 200 μm or at least about 50 μm and no more than about 100 μm), inclusive of all values and ranges therebetween. In some embodiments, the first separator **150***a* and/or the second separator **150***b* can have a thickness of about 1 μm, about 2 μm, about 3 μm, about 4 μm, about 5 μm, about 6 μm, about 7 μm, about 8 μm, about 9 μm, about 10 μm, about 20 μm, about 30 μm, about 40 μm, about 50 μm, about 60 μm, about 70 μm, about 80 μm, about 90 μm, about 100 μm, about 110 μm, about 120 μm, about 130 μm, about 140 μm, about 150 μm, about 160 μm, about 170 μm, about 180 μm, about 190 μm, or about 200 μm. In some embodiments, the first separator **150***a* can have a thickness the same or substantially similar to the thickness of the second separator **150***b*. In some embodiments, the first separator **150***a* can have a thickness greater or less than a thickness of the second separator **150***b*.

In some embodiments, the first separator **150***a* can be coupled to the second separator **150***b*. In some embodiments, the first separator **150***a* and/or the second separator **150***b* can be wetted so as to promote clinging between first separator **150***a* and the second separator **150***b*. In other words, the first separator **150***a* can be held to the second separator **150***b* via surface tension and/or capillary forces.

In some embodiments, the anode material **110**, the anode current collector **120**, and the first separator **150***a* can be packaged in a first container, while the cathode material **130**, the cathode current collector **140** and the second separator **150***b* can be packaged in a second container prior to assembly. In other words, the electrochemical cell **100** can be assembled via an anode kit (including the anode material **110**, the anode current collector **120**, and the first separator **150***a*) and a cathode kit (including the cathode material **130**, the cathode current collector **140**, and the second separator **150***b*). The anode material **110**, the anode current collector **120**, and the first separator **150***a* can be removed from the first container and the cathode material **130**, the cathode current collector **140**, and the second separator **150***b* can be removed from the second container. The first separator **150***a* can then be disposed on the second separator **150***b* to form the electrochemical cell **100**.

FIG. **2** shows an illustration of an electrochemical cell **200**, according to an embodiment. As shown, the electrochemical cell **200** includes an anode material **210** disposed on an anode current collector **220**, a cathode material **230** disposed on a cathode current collector **240**, a first separator **250***a* disposed on the anode material **210**, a second separator **250***b* disposed on the cathode material **230**, and an interlayer **260** disposed between the first separator **250***a* and the second separator **250***b*. In some embodiments, the anode material **210**, the anode current collector **220**, the cathode material **230**, the cathode current collector **240**, the first separator **250***a*, and the second separator **250***b* can be the same or substantially similar to the anode material **110**, the anode current collector **120**, the cathode material **130**, the cathode current collector **140**, the first separator **150***a*, and the second separator **150***b*, as described above with reference to FIG. **1**. Thus, certain aspects of the anode material **210**, the anode current collector **220**, the cathode material **230**, the cathode current collector **240**, the first separator **250***a*, and the second separator **250***b* are not described in greater detail herein.

In some embodiments, the interlayer **260** can include an electrolyte layer. In some embodiments, the electrolyte layer can include a liquid electrolyte. In some embodiments, the electrolyte layer can include a solid-state electrolyte, for example, to prevent dendrite growth. In some embodiments, the electrolyte layer can include polyacrylonitrile (PAN). In some embodiments, the electrolyte layer can partially or fully saturate the first separator **250***a* and/or the second separator **250***b* (collectively referred to as the separators **250**). In some embodiments, the electrolyte layer can aid in bonding the first separator **250***a* to the second separator **250***b*. In some embodiments, the electrolyte layer can create a surface tension to bond the first separator **250***a* to the second separator **250***b*. In some embodiments, the electrolyte layer can facilitate movement of electroactive species between the anode material **210** and the cathode material **230**.

In some embodiments, the interlayer **260** can have a thickness of at least about 500 nm, at least about 1 μm, at least about 2 μm, at least about 3 μm, at least about 4 μm, at least about 5 μm, at least about 6 μm, at least about 7 μm, at least about 8 μm, at least about 9 μm, at least about 10 μm, at least about 20 μm, at least about 30 μm, at least about 40 μm, at least about 50 μm, at least about 60 μm, at least about

70 μm, at least about 80 μm, or at least about 90 μm. In some embodiments, the interlayer **260** can have a thickness of no more than about 100 μm, no more than about 90 μm, no more than about 80 μm, no more than about 70 μm, no more than about 60 μm, no more than about 50 μm, no more than about 40 μm, no more than about 30 μm, no more than about 20 μm, no more than about 10 μm, no more than about 9 μm, no more than about 8 μm, no more than about 7 μm, no more than about 6 μm, no more than about 5 μm, no more than about 4 μm, no more than about 3 μm, no more than about 2 μm, or no more than about 1 μm. Combinations of the above-referenced thicknesses of the interlayer **260** are also possible (e.g., at least about 500 nm and no more than about 100 μm or at least about 2 μm and no more than about 30 μm), inclusive of all values and ranges therebetween. In some embodiments, the interlayer **260** can have a thickness of about 500 nm, about 1 μm, about 2 μm, about 3 μm, about 4 μm, about 5 μm, about 6 μm, about 7 μm, about 8 μm, about 9 μm, about 10 μm, about 20 μm, about 30 μm, about 40 μm, about 50 μm, about 60 μm, about 70 μm, about 80 μm, about 90 μm, or about 100 μm.

In some embodiments, the interlayer **260** can include a semi-solid electrode layer disposed between the first separator **250***a* and the second separator **250***b*. In some embodiments, the layer of semi-solid electrode material can be included in addition to the electrolyte layer. In some embodiments, the semi-solid electrode layer between the first separator **250***a* and the second separator **250***b* can include a material that reacts with metallic lithium. In some embodiments, the semi-solid electrode layer between the first separator **250***a* and the second separator **250***b* can include hard carbon, graphite, or any other suitable electrode material or combinations thereof. In some embodiments, if the anode material **210** begins to form dendrites that penetrate the first separator **250***a*, the dendritic material can react with the semi-solid electrode layer between the first separator **250***a* and the second separator **250***b*, such that the dendrites dissipate, thus preventing a short circuit. In some embodiments, the interlayer **260** may include a single layer. In some embodiments, the interlayer **260** may include a bilayer structure, for example, include a first layer including a semi-solid electrode layer (e.g., a binder-free carbon slurry), and a second layer including a solid state electrolyte (e.g., LLZO, LLTO, LATP, sulfides, polymer gel electrolytes, etc.).

In some embodiments, the semi-solid electrode layer between the first separator **250***a* and the second separator **250***b* can aid in transporting electroactive species across the separators **250**. The semi-solid electrode layer between the first separator **250***a* and the second separator **250***b* can provide reduced tortuosity and better lithium ion diffusion compared to conventional electrode materials. The composition of the semi-solid electrode layer between the first separator **250***a* and the second separator **250***b* can be fine-tuned to facilitate ion movement therethrough.

In some embodiments, the semi-solid electrode layer between the first separator **250***a* and the second separator **250***b* can have catalytic effects to remove, dissolve, and/or corrode contaminating metal powders (e.g., iron, chromium, nickel, aluminum, copper). In such cases, the semi-solid electrode layer between the first separator **250***a* and the second separator **250***b* can serve as a metal contamination removing buffer layer. In some embodiments, the semi-solid electrode layer between the first separator **250***a* and the second separator **250***b* can include a non-lithium ion semi-solid slurry with an aligned pore structure, a high surface area, and/or a diffusive structure combined with an electrolyte. Such materials can include metal-organic frameworks (MOFs), carbon black, an anode aluminum oxide (AAO) template, and/or silica. In such cases, the semi-solid electrode layer between the first separator **250***a* and the second separator **250***b* can serve as an electrolyte reservoir and/or an embedding base for a dendrite-removing catalyst. Such materials can also improve current distributions in the electrochemical cell **200**. In some embodiments, the dendrite-removing catalyst can include a metal base and/or a polymer base for the facilitation of redox reactions. In some embodiments, the dendrite-removing catalyst can include fluorine, sulfide, or any other suitable catalyst or combinations thereof. In some embodiments, the catalyst can include a base polymer coating mix or a carbon mix.

In some embodiments, the interlayer **260** can include a conventional (i.e., solid) electrode layer can be disposed between the first separator **250***a* and the second separator **250***b*. In some embodiments, the conventional electrode layer between the first separator **250***a* and the second separator **250***b* can include a binder (e.g., a solid binder or a gel binder).

In some embodiments, the interlayer **260** can include a polyolefin, a solid-state electrolyte sheet, and/or a polymer electrolyte sheet. In some embodiments, the interlayer **260** can include polyacrylonitrile (PAN), poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP), poly(methyl methacrylate) (PMMA), polyacrylic acid (PAA), polyethylene oxide (PEO), or any combination thereof.

In some embodiments, the interlayer **260** can include a cathode. In some embodiments, the cathode in the interlayer **260** can include lithium titanate (LTO), hard carbon (HC), and/or any other material with a high impedance connection. In some embodiments, the LTO can include an electron-conductive LTO, such as $Li_{4+x}Ti_5O_{12}$ and/or $Li_4Ti_5O_{12-x}$. In some embodiments, the interlayer **260** can include lithium iron phosphate (LFP) with a high impedance connection. The LFP can be considered a safe chemistry for the monitoring of dendrite formation. If a dendrite forms in either of the electrodes and penetrates into the interlayer **260**, the dendrite would be consumed. Also, voltage can be monitored between the interlayer **260** and the anode current collector **220**, as shown. In some embodiments, voltage can be monitored between the interlayer **260** and the cathode current collector **240**. This voltage monitoring can detect if a dendrite has reached the interlayer **260**.

Measuring voltage between the interlayer **260** and the anode current collector **220** and/or the cathode current collector **240** can be a more efficient method of detecting defects in the electrochemical cell **200** than measuring across the entire electrochemical cell **200** (i.e., between the anode current collector **220** and the cathode current collector **240**), particularly in modules with multiple cells. In some embodiments, multiple electrochemical cells can be connected in parallel and/or series to produce a cell module. For example, if a cell has a capacity of 3 Ah, 50 such cells can be connected in parallel to produce a capacity of 150 Ah. In some embodiments, the module can include about 1, about 2, about 3, about 4, about 5, about 6, about 7, about 8, about 9, about 10, about 11, about 12, about 13, about 14, about 15, about 16, about 17, about 18, about 19, about 20, about 25, about 30, about 35, about 40, about 45, about 50, about 55, about 60, about 65, about 70, about 75, about 80, about 85, about 90, about 95, or about 100 electrochemical cells connected in series and/or parallel, inclusive of all values and ranges therebetween. In monitoring the voltage across the interlayer **260** and the anode current collector **220** and/or

the cathode current collector **240**, the interlayer **260** can serve as a reference electrode.

In some embodiments, individual electrochemical cell in a module can have a capacity of about 0.5 Ah, about 1 Ah, about 2 Ah, about 3 Ah, about 4 Ah, about 5 Ah, about 6 Ah, about 7 Ah, about 8 Ah, about 9 Ah, or about 10 Ah, inclusive of all values and ranges therebetween. In some embodiments, modules described herein can have a capacity of about 10 Ah, about 20 Ah, about 30 Ah, about 40 Ah, about 50 Ah, about 60 Ah, about 70 Ah, about 80 Ah, about 90 Ah, about 100 Ah, about 110 Ah, about 120 Ah, about 130 Ah, about 140 Ah, about 150 Ah, about 160 Ah, about 170 Ah, about 180 Ah, about 190 Ah, about 200 Ah, about 210 Ah, about 220 Ah, about 230 Ah, about 240 Ah, about 250 Ah, about 260 Ah, about 270 Ah, about 280 Ah, about 290 Ah, or about 300 Ah, inclusive of all values and ranges therebetween.

In existing battery management systems (BMS) and cell modules, the ability to diagnose the health of each electrochemical cell is limited. To monitor the health of each cell, local voltage and/or current measurements are used to discern small changes in cell voltages. Voltage measurements across individual cells offer little direct correlation to the individual cell health. The addition of differential current measurement in the modules adversely affects the total system complexity and the cost of the measurement systems. Conversely, if the voltage between the interlayer **260** and a current collector is measured for each parallel electrochemical cell, the relative difference of that voltage is a direct measure of the relative health (i.e., impedance) of the anode material **210** and/or the cathode material **230** and relative impedance within the electrochemical cell **200** itself. In such an arrangement, it is possible to determine (by direct measurement) if the electrochemical cell **200** is behaving normally relative to other electrochemical cells in a parallel string or pack system. Through mass data collection, trend data from a large collection of electrochemical cells can be used to coordinate the analysis of a group or lot, or individual cell serial numbers relative to the larger cell population.

For extremely large format cells, the added complexity to measure an additional 2-3 differential voltages is lower than the added complexity of adding equivalent high gain current measurement channels, or to add hall effect type sensors, for example. In this way, individual cell health of a parallel cell grouping can be directly measured. Additional diagnostics to remaining cells connected in series can be evaluated based on state-of-charge (SOC) and state-of-health (SOH) algorithms. This allows for early notification of system failures long before faults would normally be detected. This precision can also allow for a prediction of a date of failure and advanced planning. For example, materials can be positioned properly in an electrochemical cell module in anticipation of a failure. Additionally, supply chain issues can be considered before an original equipment manufacturer (OEM) fleet or an individual consumer is notified of a fault. After the voltage measurement between the interlayer **260** and the anode current collector **220** and/or the cathode current collector **240** detects a soft short-circuit, an external short of the cell module can be triggered to discharge.

In some embodiments, the interlayer **260** can prevent dangerous short circuit events from dendrite growth via metal contamination (e.g., iron contamination, zinc contamination, copper contamination) and shuttling by a buffer layer. In such a case, an iron dendrite can grow and touch hard carbon, graphite, and/or other carbon-containing materials in the interlayer **260**, with the interlayer **260** having a cathode potential. Once the iron dendrite touches the hard carbon, graphite, and/or the other carbon-containing materials in the interlayer **260**, the iron dissolves under the cathode potential, but the high current moving through the electrochemical cell **200** persists via a connection through a diode or high resistance. When metal contamination is used to prevent dangerous short circuit events, voltage can be monitored between the interlayer **260** and the anode current collector **220** and/or the cathode current collector **240** (or between the interlayer **260** and the anode material **210** and/or the cathode material **230**). In some embodiments, additional safety actions can be triggered by a BMS if a significant voltage drop (e.g., at least about 0.5 V, at least about 1 V, at least about 1.5 V, at least about 2 V, at least about 2.5 V, at least about 3 V, at least about 3.5 V, at least about 4 V, at least about 4.5 V, at least about 5 V, inclusive of all values and ranges therebetween) is detected. In some embodiments, the interlayer **260** may include a tab to enable coupling with an electrical connection or sensing system external to the electrochemical cell **200**. In some embodiments, multiple electrochemical cells can be connected in parallel with a tab connected to the interlayer **260**. A diode or high resistance resistor can be connected to many cathodes (e.g., many tabs connected to cathode current collectors **240**) and many interlayers (e.g., many tabs connected to interlayers **260**).

In some embodiments, the interlayer **260** can prevent dangerous short circuit events from lithium dendrites via lithium intercalation. For example, lithium dendrites can grow and penetrate the first separator **250*a*** or the second separator **250*b*** and contact hard carbon, graphite, and/or a carbon-containing material in the interlayer **260**. Once the lithium dendrite contacts the hard carbon, graphite, and/or the carbon-containing material in the interlayer **260**, the lithium intercalates into the carbon, graphite, and/or the carbon-containing material. While hard carbon, graphite, and/or any carbon-containing material can facilitate lithium intercalation, any material that reacts with lithium can achieve this lithium intercalation function. In some embodiments, the interlayer **260** can include silicon, aluminum, silver, tungsten, tin, or any combination thereof.

FIG. **3** shows a block diagram of a method **10** of forming an electrochemical cell, according to an embodiment. As shown, the method **10** includes disposing an anode material onto an anode current collector at step **11** and disposing a first separator onto the anode material at step **12**. The method **10** optionally includes conveying the anode current collector, the anode material, and the first separator (collectively referred to as "the anode") in step **13**. The method **10** further includes disposing a cathode material onto a cathode current collector at step **14** and disposing a second separator onto the cathode material at step **15**. The method **10** optionally includes conveying the cathode current collector, the cathode material, and the second separator (collectively referred to as "the cathode") at step **16** and wetting the first separator and/or the second separator at step **17**. The method **10** includes coupling the first separator to the second separator at step **18** to form the electrochemical cell.

Step **11** includes disposing the anode material onto the anode current collector. The anode material and the anode current collector can have any of the properties of the anode material **110** and the anode current collector **120** (e.g., thickness, composition) as described above with reference to FIG. **1**. In some embodiments, the anode material can be extruded (e.g., via a twin-screw extruder) onto the anode current collector. In some embodiments, the anode material can be dispensed via a nozzle. In some embodiments, the

dispensation of the anode material can be via any of the methods described in U.S. provisional application 63/115,293 (hereinafter "the '293 application"), entitled, "Methods of Continuous and Semi-Continuous Production of Electrochemical Cells," filed Nov. 18, 2020, the entirety of which is incorporated herein by reference. In some embodiments, the dispensation of the anode material can be via any of the methods described in U.S. patent publication no. 2020/0014025 (hereinafter "the '025 publication), entitled "Continuous and Semi-Continuous Methods of Semi-Solid Electrode and Battery Manufacturing," filed Jul. 9, 2019, the entirety of which is hereby incorporated by reference.

Step **12** includes disposing the first separator onto the anode material. In some embodiments, the first separator can be pre-soaked or pre-coated with electrolyte solution prior to the disposing. In some embodiments, the first separator can be placed onto the anode material by a machine. In some embodiments, the first separator can be placed onto the anode material via one or more rollers, conveying separator material. In some embodiments, the separator can be placed onto the anode material via any of the methods described in the '293 application and/or the '025 publication.

Step **13** optionally includes conveying the anode. In some embodiments, the conveying can be on a conveyance device, such as a conveyor belt. In some embodiments, the conveying can be through a tunnel to limit evaporation of electrolyte solution from the anode. In some embodiments, the anode can be on the conveyance device for at least about 1 second, at least about 5 seconds, at least about 10 seconds, at least about 20 seconds, at least about 30 seconds, at least about 40 seconds, at least about 50 seconds, at least about 1 minute, at least about 5 minutes, at least about 10 minutes, at least about 20 minutes, at least about 30 minutes, at least about 40 minutes, at least about 50 minutes, at least about 1 hour, at least about 2 hours, at least about 3 hours, at least about 4 hours, at least about 5 hours, at least about 10 hours, at least about 15 hours, or at least about 20 hours. In some embodiments, the anode can be on the conveyance device for no more than about 1 day, no more than about 20 hours, no more than about 15 hours, no more than about 10 hours, no more than about 5 hours, no more than about 4 hours, no more than about 3 hours, no more than about 2 hours, no more than about 1 hour, no more than about 50 minutes, no more than about 40 minutes, no more than about 30 minutes, no more than about 20 minutes, no more than about 10 minutes, no more than about 5 minutes, no more than about 4 minutes, no more than about 3 minutes, no more than about 2 minutes, no more than about 1 minute, no more than about 50 seconds, no more than about 40 seconds, no more than about 30 seconds, no more than about 20 seconds, no more than about 10 seconds, or no more than about 5 seconds.

Combinations of the above-referenced time periods the anode remains on the conveyance device are also possible (e.g., at least about 1 second and no more than about 1 day or at least about 5 minutes and no more than about 2 hours), inclusive of all values and ranges therebetween. In some embodiments, the anode can be on the conveyance device for about 1 second, about 5 seconds, about 10 seconds, about 20 seconds, about 30 seconds, about 40 seconds, about 50 seconds, about 1 minute, about 5 minutes, about 10 minutes, about 20 minutes, about 30 minutes, about 40 minutes, about 50 minutes, about 1 hour, about 2 hours, about 3 hours, about 4 hours, about 5 hours, about 10 hours, about 15 hours, about 20 hours, or about 1 day.

In some embodiments, the anode can be conveyed a distance of at least about 1 cm, at least about 2 cm, at least about 3 cm, at least about 4 cm, at least about 5 cm, at least about 10 cm, at least about 20 cm, at least about 30 cm, at least about 40 cm, at least about 50 cm, at least about 60 cm, at least about 70 cm, at least about 80 cm, at least about 90 cm, at least about 1 m, at least about 2 m, at least about 3 m, at least about 4 m, at least about 5 m, at least about 6 m, at least about 7 m, at least about 8 m, at least about 9 m, at least about 10 m, at least about 20 m, at least about 30 m, at least about 40 m, at least about 50 m, at least about 60 m, at least about 70 m, at least about 80 m, or at least about 90 m. In some embodiments, the anode can be conveyed a distance of no more than about 100 m, no more than about 90 m, no more than about 80 m, no more than about 70 m, no more than about 60 m, no more than about 50 m, no more than about 40 m, no more than about 30 m, no more than about 20 m, no more than about 10 m, no more than about 9 m, no more than about 8 m, no more than about 7 m, no more than about 6 m, no more than about 5 m, no more than about 4 m, no more than about 3 m, no more than about 2 m, no more than about 1 m, no more than about 90 cm, no more than about 80 cm, no more than about 70 cm, no more than about 60 cm, no more than about 50 cm, no more than about 40 cm, no more than about 30 cm, no more than about 20 cm, no more than about 10 cm, no more than about 9 cm, no more than about 8 cm, no more than about 7 cm, no more than about 6 cm, no more than about 5 cm, no more than about 4 cm, no more than about 3 cm, or no more than about 2 cm. Combinations of the above-referenced conveyance distances are also possible (e.g., at least about 1 cm and no more than about 100 m or at least about 50 cm and no more than about 20 m), inclusive of all values and ranges therebetween. In some embodiments, the anode can be conveyed about 1 cm, about 2 cm, about 3 cm, about 4 cm, about 5 cm, about 10 cm, about 20 cm, about 30 cm, about 40 cm, about 50 cm, about 60 cm, about 70 cm, about 80 cm, about 90 cm, about 1 m, about 2 m, about 3 m, about 4 m, about 5 m, about 6 m, about 7 m, about 8 m, about 9 m, about 10 m, about 20 m, about 30 m, about 40 m, about 50 m, about 60 m, about 70 m, about 80 m, about 90 m, or about 100 m.

The separator coating the anode material can prevent electrolyte from evaporating during conveyance or other portions of the production process. By covering the surface of the anode material distal to the anode current collector a significant percentage of the surface of the anode material (e.g., 90-95%) is not exposed to the surrounding environment. Thus, a significant portion of the avenues for evaporation are restricted.

Step **14** includes disposing the cathode material onto the cathode current collector. In some embodiments, the dispensation of the cathode material can have the same or substantially similar properties to those described above with reference to the anode material in step **11**. Step **15** includes disposing the second separator onto the cathode material. In some embodiments, the disposal of the second separator onto the cathode material can have the same or substantially similar properties to those described above with reference to the first separator in step **12**.

Step **16** optionally includes conveying the cathode. In some embodiments, the duration and distance of the conveying of the cathode can be the same or substantially similar to the duration and distance of the conveying of the anode with reference to step **13**. In some embodiments, the cathode can be conveyed on the same conveyor as the anode. In some embodiments, the anode can be conveyed on a first conveyor and the cathode can be conveyed on a second conveyor, the second conveyor different from the first conveyor.

Step **17** optionally includes wetting the first separator and/or the second separator. The wetting can be via a wetting agent. In some embodiments, the wetting agent can include an electrolyte solvent without electrolyte salt. In some embodiments, the wetting agent can include an electrolyte solution. In some embodiments, the wetting agent can include a diluted electrolyte solution (i.e., an electrolyte solution with a salt concentration lower than the targeted salt concentration in the finished electrochemical cell). In some embodiments, the wetting agent can have an electrolyte salt concentration of at least about 0.1 M, at least about 0.2 M, at least about 0.3 M, at least about 0.4 M, at least about 0.5 M, at least about 0.6 M, at least about 0.7 M, at least about 0.8 M, or at least about 0.9 M. In some embodiments, the wetting agent can have an electrolyte salt concentration of no more than about 1 M, no more than about 0.9 M, no more than about 0.8 M, no more than about 0.7 M, no more than about 0.6 M, no more than about 0.5 M, no more than about 0.4 M, no more than about 0.3 M, no more than about 0.2 M. Combinations of the above-referenced concentrations of electrolyte salt in the wetting agent are also possible (e.g., at least about 0.1 M and no more than about 1 M or at least about 0.4 M and no more than about 0.6 M), inclusive of all values and ranges therebetween. In some embodiments, the wetting agent can have an electrolyte salt concentration of about 0.1 M, about 0.2 M, about 0.3 M, about 0.4 M, about 0.5 M, about 0.6 M, about 0.7 M, about 0.8 M, about 0.9 M, or about 1 M.

In some embodiments, the electrolyte solvent can include ethyl methyl carbonate (EMC), ethylene carbonate (EC), propylene carbonate (PC), dimethyl carbonate (DMC), butylene carbonate, and their chlorinated or fluorinated derivatives, and/or a family of acyclic dialkyl carbonate esters, such as dimethyl carbonate, diethyl carbonate, ethylmethyl carbonate, dipropyl carbonate, methyl propyl carbonate, ethyl propyl carbonate, dibutyl carbonate, butylmethyl carbonate, butylethyl carbonate and butylpropyl carbonate. In some embodiments, the electrolyte solvent can include gamma-Butyrolactone (GBL), dimethoxyethane, tetrahydrofuran, 2-methyl tetrahydrofuran, 1,3-dioxolane, 1,4-dioxane, 4-methyl-1,3-dioxolane, diethyl ether, sulfolane, methylsulfolane, acetonitrile, propiononitrile, ethyl acetate, methyl propionate, ethyl propionate, dimethyl carbonate, tetraglyme, monoglyme, dioxane, or any other suitable electrolyte solvent. In some embodiments, the electrolyte salt can include $LiClO_4$, $LiPF_6$, $LiBF_4$, LiTFSI, LiBETI, LiBOB, Lithium difluoro(oxalato)borate (LIODFB), Lithium bis(fluorosulfonyl)imide (LiFSI), or any other appropriate electrolyte salt.

In some embodiments, the wetting agent can be sprayed onto the first separator and/or the second separator. In some embodiments, the wetting agent can be brushed onto the first separator and/or the separator. In some embodiments, the wetting agent can be applied to the first separator via a first method and the second separator via a second method, the second method different from the first method. In some embodiments, the wetting agent can aid in adhering the separators to their respective electrode materials.

In some embodiments, less than about 20%, less than about 19%, less than about 18%, less than about 17%, less than about 16%, less than about 15%, less than about 14%, less than about 13%, less than about 12%, less than about 11%, less than about 10%, less than about 9%, less than about 8%, less than about 7%, less than about 6%, less than about 5%, less than about 4%, less than about 3%, less than about 2%, or less than about 1% by volume of the wetting agent can be lost to evaporation during execution of the method **10**.

In some embodiments, less than about 20%, less than about 19%, less than about 18%, less than about 17%, less than about 16%, less than about 15%, less than about 14%, less than about 13%, less than about 12%, less than about 11%, less than about 10%, less than about 9%, less than about 8%, less than about 7%, less than about 6%, less than about 5%, less than about 4%, less than about 3%, less than about 2%, or less than about 1% by volume of the electrolyte solution in the anode material can be lost to evaporation during execution of the method.

In some embodiments, less than about 20%, less than about 19%, less than about 18%, less than about 17%, less than about 16%, less than about 15%, less than about 14%, less than about 13%, less than about 12%, less than about 11%, less than about 10%, less than about 9%, less than about 8%, less than about 7%, less than about 6%, less than about 5%, less than about 4%, less than about 3%, less than about 2%, or less than about 1% by volume of the electrolyte solution in the cathode material can be lost to evaporation during execution of the method.

In some embodiments, less than about 20%, less than about 19%, less than about 18%, less than about 17%, less than about 16%, less than about 15%, less than about 14%, less than about 13%, less than about 12%, less than about 11%, less than about 10%, less than about 9%, less than about 8%, less than about 7%, less than about 6%, less than about 5%, less than about 4%, less than about 3%, less than about 2%, or less than about 1% by volume of the combination of the electrolyte solution in the anode material, the electrolyte solution in the cathode material, and the wetting agent can be lost to evaporation during execution of the method.

In some embodiments, the anode material, the anode current collector, and the first separator can be disposed in a first package as an anode kit. In some embodiments, the cathode material, the cathode current collector, and the second separator can be disposed in a second package as a cathode kit. In some embodiments, the first separator can be wetted prior to disposal in the first package. In some embodiments, the second separator can be wetted prior to disposal in the second package.

Step **18** includes coupling the first separator to the second separator to form the electrochemical cell. In some embodiments, the coupling can include adhering the first separator to the second separator. In some embodiments, the first separator and/or the second separator can be wetted to facilitate the adhering of the first separator to the second separator. In some embodiments, the anode material, the anode current collector, and the first separator can be removed from the first package prior to coupling the first separator to the second separator to form the electrochemical cell. In some embodiments, the cathode material, the cathode current collector, and the second separator can be removed from the second package prior to coupling the first separator to the second separator. In some embodiments, a semi-solid electrode material can be applied to the first separator and/or the second separator prior to coupling the first separator to the second separator, such that the semi-solid electrode material is disposed between the first separator and the second separator in the electrochemical cell. In some embodiments, a conventional electrode material can be applied to the first separator and/or the second separator prior to coupling the first separator to the second separator, such that the conventional electrode material is disposed

between the first separator and the second separator in the electrochemical cell. In some embodiments, a solid-state electrolyte material can be applied to the first separator and/or the second separator prior to coupling the first separator to the second separator, such that the solid-state electrolyte material is disposed between the first separator and the second separator in the electrochemical cell.

FIGS. **4A-4C** show conveyance of electrodes, according to various embodiments. FIGS. **4A-4C** show an anode material **410** disposed on an anode current collector **420** and being conveyed by a first conveyor **90***a*, and a cathode material **430** disposed on a cathode current collector **440** and being conveyed by a second conveyor **90***b*. FIG. **4A** shows the electrodes without any separators disposed thereon. As shown in FIG. **4A**, electrolyte solvent ES evaporates from both of the electrodes. FIG. **4B** shows a first separator **450***a* disposed on the anode material **410**. As shown in FIG. **4B**, electrolyte solvent ES evaporates from the cathode material **430** but evaporation of electrolyte solvent ES from the anode material **410** is eliminated or significantly reduced. FIG. **4B** shows the first separator **450***a* disposed on the anode material **410** and a second separator **450***b* disposed on the cathode material **430**. As shown in FIG. **4C**, evaporation of electrolyte solvent ES from both the anode material **410** and the cathode material **430** is eliminated or significantly reduced. As shown, the first conveyor **90***a* is a separate conveyor from the second conveyor **90***b*. In some embodiments, the anode material **410**, the anode current collector **420**, the cathode material **430**, and the cathode current collector **440** can all be conveyed on the same conveyor.

FIG. **5** shows an illustration of an electrochemical cell **500**, according to an embodiment. The electrochemical cell **500** is substantially similar to the electrochemical cell **200**, and includes an anode material **510** disposed on an anode current collector **520**, a cathode material **530** disposed on a cathode current collector **540**, a first separator **550***a* disposed on the anode material **510**, and a second separator **550***b* disposed on the cathode material **530**, which can be substantially similar to the anode material **210**, the anode current collector **220**, the cathode material **230**, the cathode current collector **240**, the first separator **250***a*, and the second separator **250***b*, respectively, as previously described herein with respect to the electrochemical cell **200**.

However, different from electrochemical cell **200**, the electrochemical cell **500** also includes a third separator **550***c* disposed between the first separator **550***a* and the second separator **550***b*. A first interlayer **560***a* is disposed between the first separator **550***a* and the third separator **550***c*, and a second interlayer **560***b* is disposed between the second separator **550***b* and the third separator **550***c*. The third separator **550***c* may be substantially similar to the first separator **250***a* and/or the second separator **250***b* as described in detail with respect to the electrochemical cell **200**.

In some embodiments, the first interlayer **560***a* and/or the second interlayer **560***b* can include an electrolyte layer, as described in detail with respect to FIG.**2**. In some embodiments, the electrolyte layer can partially or fully saturate the first separator **550***a*, the second separator **550***b*, and or the third separator **550** (collectively referred to as the separators **550**). In some embodiments, the electrolyte layer can aid in bonding the first separator **550***a* and second separator **550***b* to the third separator **550***c*. In some embodiments, the electrolyte layer can create a surface tension to bond the first separator **550***a* and the second separator **550***b* to the third separator **550***c*. In some embodiments, the electrolyte layer can facilitate movement of electroactive species between the anode material **510** and the cathode material **530**.

In some embodiments, the first interlayer **560***a* and/or the second interlayer **560***b* can have any thickness as described in detail with respect to the interlayer **260** of the electrochemical cell **200**. In some embodiments, the first interlayer **560***a* and the second interlayer can include a semi-solid electrode layer disposed between the first separator **550***a* and third separator **550***c*, and the second separator **550***b* and the third separator **550***c*, respectively. In some embodiments, the layer of semi-solid electrode material can be included in addition to the electrolyte layer. In some embodiments, the semi-solid electrode layer between the first separator **550***a* and the third separator **550***c*, and/or between the second separator **550***b* and the third separator **550***c* can include a material that reacts with metallic lithium. In some embodiments, the semi-solid electrode layer between the first separator **550***a* and the third separator **550***c*, and/or between the second separator **550***b* and the third separator **550***c* can include hard carbon, graphite, or any other suitable electrode material or combinations thereof. In some embodiments, if the anode material **510** begins to form dendrites that penetrate the first separator **550***a*, the dendritic material can react with the semi-solid electrode layer between the first separator **550***a* and the third separator **550***b*, such that the dendrites dissipate, thus preventing a short circuit. In some embodiments, one or more of the first separator **550***a*, the second separator **550***b*, and the third separator **550***c* may include solid-state electrolyte sheets.

In some embodiments, the semi-solid electrode layer can aid in transporting electroactive species across the separators **550**, and/or provide reduced tortuosity and better lithium ion diffusion compared to conventional electrode materials. The composition of the semi-solid electrode layer(s) can be fine-tuned to facilitate ion movement therethrough. In some embodiments, the semi-solid electrode layer between the first separator **550***a* and third separator **550***c*, and/or between the second separator **550***b* and the third separator **550***c* can have catalytic effects to remove, dissolve, and/or corrode contaminating metal powders (e.g., iron, chromium, nickel, aluminum, copper). In such cases, the semi-solid electrode layer(s) can serve as a metal contamination removing buffer layer. In some embodiments, the semi-solid electrode layer(s) can include a non-lithium ion semi-solid slurry with an aligned pore structure, a high surface area, and/or a diffusive structure combined with an electrolyte. Such materials can include metal-organic frameworks (MOFs), carbon black, an anode aluminum oxide (AAO) template, and/or silica. In such cases, the semi-solid electrode layer(s) can serve as an electrolyte reservoir and/or an embedding base for a dendrite-removing catalyst. Such materials can also improve current distributions in the electrochemical cell **250**. In some embodiments, the dendrite-removing catalyst can include a metal base and/or a polymer base for the facilitation of redox reactions. In some embodiments, the dendrite-removing catalyst can include fluorine, sulfide, or any other suitable catalyst or combinations thereof. In some embodiments, the catalyst can include a base polymer coating mix or a carbon mix.

In some embodiments, the interlayers **560***a* and/or **560***b* can include a conventional (i.e., solid) electrode layer disposed between the first separator **550***a* and the third separator **550***c*, and/or the second separator **550***b* and the third separator **550***c*, respectively. In some embodiments, the conventional electrode layer between the respective separators **550** can include a binder (e.g., a solid binder or a gel binder).

In some embodiments, the interlayers **560***a* and/or **560***b* can include a polyolefin, a solid-state electrolyte sheet, and/or a polymer electrolyte sheet. In some embodiments, the interlayers **560***a* and/or **560***b* can include polyacrylonitrile (PAN), poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP), poly(methyl methacrylate) (PMMA), polyacrylic acid (PAA), polyethylene oxide (PEO), or any combination thereof.

In some embodiments, the interlayers **560***a* and/or **560***b* can include a cathode. In some embodiments, the cathode in the interlayers **560***a* and/or **560***b* can include lithium titanate (LTO), hard carbon (HC), and/or any other material with a high impedance connection. In some embodiments, the LTO can include an electron-conductive LTO, such as $Li_{4+x}Ti_5O_{12}$ and/or $Li_4Ti_5O_{12-x}$. In some embodiments, the interlayers **560***a* and/or **560***b* can include lithium iron phosphate (LFP) with a high impedance connection. The LFP can be considered a safe chemistry for the monitoring of dendrite formation. If a dendrite forms in either of the electrodes and penetrates into the interlayers **560***a* and/or **560***b*, the dendrite would be consumed. Also, voltage can be monitored between the first interlayer **560***a* and the anode current collector **520**, as shown. In some embodiments, voltage can be monitored between the second interlayer **560***b* and the cathode current collector **540**. This voltage monitoring can detect if a dendrite has reached the interlayers **560***a* and/or **560***b*.

Measuring voltage between the first interlayer **560***a* and the anode current collector **520** and/or the second interlayer **560***b* and the cathode current collector **540** can be a more efficient method of detecting defects in the electrochemical cell **500** than measuring across the entire electrochemical cell **500** (i.e., between the anode current collector **520** and the cathode current collector **540**), particularly in modules with multiple cells. In some embodiments, multiple electrochemical cells can be connected in parallel and/or series to produce a cell module. For example, if a cell has a capacity of 3 Ah, 50 such cells can be connected in parallel to produce a capacity of 150 Ah. In some embodiments, the module can include about 1, about 2, about 3, about 4, about 5, about 6, about 7, about 8, about 9, about 10, about 11, about 12, about 13, about 14, about 15, about 16, about 17, about 18, about 19, about 20, about 25, about 30, about 35, about 40, about 45, about 50, about 55, about 60, about 65, about 70, about 75, about 80, about 85, about 90, about 95, or about 100 electrochemical cells connected in series and/or parallel, inclusive of all values and ranges therebetween. In monitoring the voltage across the first interlayer **560***a* and the anode current collector **520** and/or the second interlayer **560***b* and the cathode current collector **540**, the first interlayer **560***a* and/or the second interlayer **560***b* can serve as a reference electrode.

In some embodiments, individual electrochemical cell in a module can have a capacity of about 0.5 Ah, about 1 Ah, about 2 Ah, about 3 Ah, about 4 Ah, about 5 Ah, about 6 Ah, about 7 Ah, about 8 Ah, about 9 Ah, or about 10 Ah, inclusive of all values and ranges therebetween. In some embodiments, modules described herein can have a capacity of about 10 Ah, about 20 Ah, about 30 Ah, about 40 Ah, about 50 Ah, about 60 Ah, about 70 Ah, about 80 Ah, about 90 Ah, about 100 Ah, about 110 Ah, about 120 Ah, about 130 Ah, about 140 Ah, about 150 Ah, about 160 Ah, about 170 Ah, about 180 Ah, about 190 Ah, about 200 Ah, about 210 Ah, about 220 Ah, about 230 Ah, about 240 Ah, about 250 Ah, about 260 Ah, about 270 Ah, about 280 Ah, about 290 Ah, or about 300 Ah, inclusive of all values and ranges therebetween.

In some embodiments, only one of the first interlayer **560***a* and the second interlayer **560***b* may be connected or coupled to an electrical source or sink. For example, in some embodiments, the first interlayer **560***a* (e.g., a graphite layer, a hard carbon layer, or any other material described herein) may not be connected with cathode, neutral, or ground, while the second interlayer **560***b* (e.g., a graphite layer, a hard carbon layer, or any other material described herein) may be coupled or connected to a cathode or neutral (e.g., via a diode or high resistance). In some embodiments, a current collector of any other layer may extend between the first separator **550***a* and the second separator **550***b* to serve as the third separator **550***c*, any may be employed as a shutdown separator. In some embodiments, the current collector of the other layer may be formed from aluminum, gold, platinum, stainless steel, titanium foil, conductive ink, etc. In some embodiments, the current collector may be formed or processed by lamination, printing (e.g., inkjet printing, gravure printing, screen printing, etc.), sputtering, spray coating, or deposition, or any other suitable method on the separator or any other suitable method. In some embodiments, a tab may be coupled to the current collector that forms one of the interlayers **560***a/b*, or directly to the interlayer **560***a* or **560***b* that does include a current collector (e.g., a graphite or hard carbon layer). In other embodiments, the tab may be formed monolithically with the current collector or otherwise interlayer **560** having a portion thereof disposed outside the electrochemical cell **500**.

FIG. **6** shows an illustration of an electrochemical cell **600**, according to an embodiment. The electrochemical cell **600** is substantially similar to the electrochemical cell **200**, and includes an anode material **610** disposed on an anode current collector **620**, a cathode material **630** disposed on a cathode current collector **640**, a first separator **650***a* disposed on the anode material **610**, a second separator **650***b* disposed on the cathode material **630**, and an interlayer **660** disposed between the first separator **650***a* and the second separator **650***b*. In some embodiments, the anode material **610**, the anode current collector **620**, the cathode material **630**, the cathode current collector **640**, and the interlayer **660** can be the same or substantially similar to the anode material **210**, the anode current collector **220**, the cathode material **230**, the cathode current collector **240**, and the interlayer **260**, as described above with reference to FIG. **2**. Thus, certain aspects of the anode material **610**, the anode current collector **620**, the cathode material **630**, the cathode current collector **640**, and the interlayer **660** are not described in greater detail herein.

Different from the electrochemical cell **200**, the first separator **650***a* and/or the second separator **650***b* may include multiple layers. For example, as shown in FIG. **6**, the first separator **650***a* includes a first separator first layer **650***a***1** and a first separator second layer **650***a***2** that are formed from different materials (e.g., any of the materials described with respect to the separators **260***a* and **260***b*). Similarly, the second separator **650***b* includes a second separator first layer **650***b***1** and a second separator second layer **650***b***2** that are formed from different materials (e.g., any of the materials described with respect to the separators **260***a* and **260***b*). The first separator second layer **650***a***2** and the second separator second layer **650***b***2** is disposed proximate to the interlayer **660** such that the interlayer **660** is interposed between the first separator second layer **650***a***2** and the second separator second layer **650***b***2**. The interlayer **660** may include a semisolid interlayer, may include a binder, or may include any other interlayer as described with respect to the interlayer **260**. In some embodiments, the first

separator first layer **650***a***1** and the second separator first layer **650***b***1** may be formed from polypropylene. In some embodiments, the first separator second layer **650***a***2** and the second separator second layer **650***b***2** may be formed from polyethylene. In other embodiments, the first separator first and/or second layers **650***a***1** and **650***a***2**, and/or the second separator first and second layers **650***b***1** and/or **650***b***2** may be formed from any other material as described herein. In some embodiments, axial end regions **670** of the first separator **650***a* and/or the second separator **650***b* that extend beyond an axial extent of the interlayer **660** may be bonded, adhered, welded, or otherwise coupled to each other so as to form a sealed pocket or cavity within which the interlayer **660** is disposed. This may advantageously prevent a semisolid or slurry based interlayer **660** from leaking from between the first separator **650***a* and the second separator **650***b*.

FIG. **7** shows an illustration of an electrochemical cell **700**, according to an embodiment. The electrochemical cell **700** is substantially similar to the electrochemical cell **200**, and includes an anode material **710** disposed on an anode current collector **720**, a cathode material **730** disposed on a cathode current collector **740**, a first separator **750***a* disposed on the anode material **710**, a second separator **750***b* disposed on the cathode material **730**, and an interlayer **760** disposed between the first separator **750***a* and the second separator **750***b*. In some embodiments, the anode material **710**, the anode current collector **720**, the cathode material **730**, the cathode current collector **740**, the first separator **750***a* and the second separator **750***b* can be the same or substantially similar to the anode material **210**, the anode current collector **220**, the cathode material **230**, the cathode current collector **240**, the first current collector **250***a*, and the second collector **250***b*, as described above with reference to FIG. **2**. Thus, certain aspects of the anode material **710**, the anode current collector **720**, the cathode material **730**, the cathode current collector **740**, the first separator **750***a*, and the second separator **750***b* are not described in greater detail herein.

In some embodiment, the interlayer **760** is a multilayer structure. For example, as shown in FIG. **7**, the interlayer **760** includes a first layer **760***a* that may include a solid-state electrolyte, for example, any of the solid-state electrolytes as described with respect to FIG. **2**. The interlayer **760** may optionally, also include a second layer **760***b* disposed between the first layer **760***a* and the first separator **750***a*, and a third layer **760***c* disposed between the first layer **760***a* and the second separator **750***b*. In some embodiments, the second layer **760***b* and/or the third layer **760***c* may include a cathode, for example, LTO, hard carbon, LFP, and/or any other material with a high impedance connection, as described in detail with respect to FIG. **2**. For example, the second layer **760***b* may include hard carbon, and the third layer **760***c* may include LFP. The second layer **760***b* may be coupled to the anode current collector **720** via a first discharge protection component **780***a* (e.g., a diode, a high resistance resistor, or any other suitable structure) and the third layer **760***c* may be coupled to the cathode current collector **740** via a second discharge protection component **780***b* (e.g., a diode, a high resistance resistor, or any other suitable structure). As previously described, an iron dendrite can grow and touch hard carbon, graphite, and/or other carbon-containing materials in the interlayer **760**, with the interlayer **760** having an anode potential in the second layer **760***b* and a cathode potential in the third layer **760***c*. Once the iron dendrite touches the hard carbon, graphite, and/or the other carbon-containing materials in the interlayer **760**, the iron dissolves under the cathode potential, but the high current moving through the electrochemical cell **700** persists via a connection through the discharge protection components **780***a* and **780***b*. When metal contamination is used to prevent dangerous short circuit events, voltage can be monitored between the interlayer **760** and the anode current collector **720** and/or the cathode current collector **740**, as described with respect to FIG. **2**.

EXAMPLES

Comparative Example 1: A lithium-copper cell was constructed (referred to herein as "Comp Ex 1"). Lithium was plated onto copper foil. The cell was cycled at 1 mA/cm^2 for 1 hour at 25° C. under a pressure of 200 psig. Stripping of the copper foil continued until a 1V cut-off was reached. A standard separator including a single layer (i.e., a single separator) was placed between the lithium foil anode and the copper foil cathode.

Example 1: A lithium-copper cell was constructed (referred to herein as "Ex 1"). Lithium was plated onto copper foil. The cell was cycled at 1 mA/cm^2 for 1 hour at 25° C. under a pressure of 200 psig. Stripping of the copper foil continued until a 1V cut-off was reached. A separator coated with a layer of hard carbon (Kuraray HC, 2-3 μm) was coupled to a standard separator and placed between the lithium foil anode and the copper foil cathode, the hard carbon layer facing the anode. The hard carbon coated separator was implemented as a means of lithiating and/or complexing with any dendritic lithium that protrudes from the lithium foil surface, enabling a safety feature to prevent a hard short from lithium plating on the anode.

Both Comp Ex 1 and Ex 1 were subject to 32 cycles before deconstruction, where lithium was stripped from the copper foil, allowing lithium to complex with the hard carbon in the Ex 1 cell. The hard carbon in the active area of lithium appeared shiny, indicating lithiation of the hard carbon. The hard carbon that was outside of the lithium foil contact area was dull in color, suggesting it was unlithiated. The cycled lithium foil anode also appeared pristine in nature, indicating the lithium was being plated on the hard carbon. No mossy or dendritic lithium was observed on the foil. FIG. **8**A-**8**B show supporting voltage profiles highlighting the difference in overpotential of Comp Ex 1 (FIG. **8**A) vs. Ex 1 (FIG. **8**B). As shown, Ex 1 with a hard carbon layer between two separators, experiences less significant overpotential losses than Comp Ex 1.

Comparative Example 2: A lithium copper cell was constructed (referred to herein as "Comp. Ex. 2") and operated similar to Comp. Ex. 1, except that Comp. Ex. 2 was cycled at 7.5 mA/cm^2 with 75% lithium ion usage.

Example 2: A lithium copper cell was constructed (referred to herein as "Ex. 2") and operated similar to Ex. 1, except that Ex. 2 was cycled at 7.5 mA/cm^2 with 75% lithium ion usage. FIGS. **9**A and **9**B show supporting voltage profiles highlighting the difference in overpotential of Comp. Ex. 2 and Ex. 2. The Comp. Ex. 2 cell that did not include the interlayer shorted in 11 cycles within 22 hours, while the Ex. 2 cell that included the hard carbon interlayer continued to operate normally after 25 cycles with minimum over polarization. This indicates that the interlayer smooths or levels the current distribution for fast lithium plating, storing, and preventing the dendrite from projecting through the separators.

As used in this specification, the singular forms "a," "an," and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, the term "a member" is intended to mean a single member or a combination of

members, "a material" is intended to mean one or more materials, or a combination thereof.

The term "substantially" when used in connection with "cylindrical," "linear," and/or other geometric relationships is intended to convey that the structure so defined is nominally cylindrical, linear or the like. As one example, a portion of a support member that is described as being "substantially linear" is intended to convey that, although linearity of the portion is desirable, some non-linearity can occur in a "substantially linear" portion. Such non-linearity can result from manufacturing tolerances, or other practical considerations (such as, for example, the pressure or force applied to the support member). Thus, a geometric construction modified by the term "substantially" includes such geometric properties within a tolerance of plus or minus 5% of the stated geometric construction. For example, a "substantially linear" portion is a portion that defines an axis or center line that is within plus or minus 5% of being linear.

As used herein, the term "set" and "plurality" can refer to multiple features or a singular feature with multiple parts. For example, when referring to a set of electrodes, the set of electrodes can be considered as one electrode with multiple portions, or the set of electrodes can be considered as multiple, distinct electrodes. Additionally, for example, when referring to a plurality of electrochemical cells, the plurality of electrochemical cells can be considered as multiple, distinct electrochemical cells or as one electrochemical cell with multiple portions. Thus, a set of portions or a plurality of portions may include multiple portions that are either continuous or discontinuous from each other. A plurality of particles or a plurality of materials can also be fabricated from multiple items that are produced separately and are later joined together (e.g., via mixing, an adhesive, or any suitable method).

As used herein, the term "semi-solid" refers to a material that is a mixture of liquid and solid phases, for example, such as a particle suspension, a slurry, a colloidal suspension, an emulsion, a gel, or a micelle.

Various concepts may be embodied as one or more methods, of which at least one example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments. Put differently, it is to be understood that such features may not necessarily be limited to a particular order of execution, but rather, any number of threads, processes, services, servers, and/or the like that may execute serially, asynchronously, concurrently, in parallel, simultaneously, synchronously, and/or the like in a manner consistent with the disclosure. As such, some of these features may be mutually contradictory, in that they cannot be simultaneously present in a single embodiment. Similarly, some features are applicable to one aspect of the innovations, and inapplicable to others.

In addition, the disclosure may include other innovations not presently described. Applicant reserves all rights in such innovations, including the right to embodiment such innovations, file additional applications, continuations, continuations-in-part, divisionals, and/or the like thereof. As such, it should be understood that advantages, embodiments, examples, functional, features, logical, operational, organizational, structural, topological, and/or other aspects of the disclosure are not to be considered limitations on the disclosure as defined by the embodiments or limitations on equivalents to the embodiments. Depending on the particular desires and/or characteristics of an individual and/or enterprise user, database configuration and/or relational model, data type, data transmission and/or network framework, syntax structure, and/or the like, various embodiments of the technology disclosed herein may be implemented in a manner that enables a great deal of flexibility and customization as described herein.

All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

As used herein, in particular embodiments, the terms "about" or "approximately" when preceding a numerical value indicates the value plus or minus a range of 10%. Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range is encompassed within the disclosure. That the upper and lower limits of these smaller ranges can independently be included in the smaller ranges is also encompassed within the disclosure, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the disclosure.

The phrase "and/or," as used herein in the specification and in the embodiments, should be understood to mean "either or both" of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with "and/or" should be construed in the same fashion, i.e., "one or more" of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the "and/or" clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to "A and/or B", when used in conjunction with open-ended language such as "comprising" can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

As used herein in the specification and in the embodiments, "or" should be understood to have the same meaning as "and/or" as defined above. For example, when separating items in a list, "or" or "and/or" shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as "only one of" or "exactly one of," or, when used in the embodiments, "consisting of," will refer to the inclusion of exactly one element of a number or list of elements. In general, the term "or" as used herein shall only be interpreted as indicating exclusive alternatives (i.e., "one or the other but not both") when preceded by terms of exclusivity, such as "either," "one of," "only one of," or "exactly one of." "Consisting essentially of," when used in the embodiments, shall have its ordinary meaning as used in the field of patent law.

As used herein in the specification and in the embodiments, the phrase "at least one," in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in

the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase "at least one" refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, "at least one of A and B" (or, equivalently, "at least one of A or B," or, equivalently "at least one of A and/or B") can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

In the embodiments, as well as in the specification above, all transitional phrases such as "comprising," "including," "carrying," "having," "containing," "involving," "holding," "composed of," and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases "consisting of" and "consisting essentially of" shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures, Section 2111.03.

While specific embodiments of the present disclosure have been outlined above, many alternatives, modifications, and variations will be apparent to those skilled in the art. Accordingly, the embodiments set forth herein are intended to be illustrative, not limiting. Various changes may be made without departing from the spirit and scope of the disclosure. Where methods and steps described above indicate certain events occurring in a certain order, those of ordinary skill in the art having the benefit of this disclosure would recognize that the ordering of certain steps may be modified and such modification are in accordance with the variations of the invention. Additionally, certain of the steps may be performed concurrently in a parallel process when possible, as well as performed sequentially as described above. The embodiments have been particularly shown and described, but it will be understood that various changes in form and details may be made.

The invention claimed is:

1. An electrochemical cell, comprising:

a first electrode disposed on a first current collector;

a second electrode disposed on a second current collector;

a first separator disposed on the first electrode;

a second separator disposed on the second electrode; and

an interlayer disposed between the first separator and the second separator, the interlayer including a first layer including a hard carbon material and a second layer different from the first layer, the second layer including a semi-solid electrode, the interlayer configured to inhibit dendrite growth and enable voltage measurement between the first electrode and the interlayer during operation of the electrochemical cell.

2. The electrochemical cell of claim **1**, further comprising:

a first electrode tab extending from the first current collector; and

an interlayer tab extending from the interlayer, the interlayer tab extending beyond the first separator and the second separator.

3. The electrochemical cell of claim **1**, wherein the hard carbon material includes a hard carbon powder.

4. The electrochemical cell of claim **1**, wherein the interlayer is configured to prevent the projection of dendrites through at least one of the first separator or the second separator during operation of the electrochemical cell.

5. The electrochemical cell of claim **1**, wherein the interlayer is configured to react with any dendritic material that has penetrated at least one of the first separator or the second separator such that the dendritic material dissipates, thus preventing a short circuit.

6. The electrochemical cell of claim **1**, wherein the first separator and the second separator are coated on the interlayer.

7. The electrochemical cell of claim **1**, wherein the first electrode comprises an electroactive species, the electroactive species including is at least one of lithium ions, sodium ions or another non-lithium ion.

8. An electrochemical cell, comprising:

a first electrode disposed on a first current collector;

a second electrode disposed on a second current collector;

a first separator disposed on the first electrode;

a second separator disposed on the second electrode;

an interlayer disposed between the first separator and the second separator, the interlayer configured to inhibit dendrite growth during operation of the electrochemical cell, the interlayer including a first layer comprising a semi-solid electrode material and a second layer different from the first layer, the second layer comprising a solid state electrolyte material;

a first electrode tab extending from the first current collector; and

a current collector material disposed between the first layer and the second layer of the interlayer, the current collector material extending from the interlayer, the current collector material extending beyond the first separator and the second separator.

9. The electrochemical cell of claim **8**, further comprising an interlayer tab extending from the interlayer, the interlayer tab including a carbonaceous material.

10. The electrochemical cell of claim **9**, wherein the carbonaceous material includes at least one of graphite, hard carbon, or carbon black.

11. The electrochemical cell of claim **8**, wherein the interlayer includes a lithium intercalation material.

12. The electrochemical cell of claim **8**, wherein the interlayer includes an electron-conductive powder.

13. An electrochemical cell, comprising:

a first electrode disposed on a first current collector;

a second electrode disposed on a second current collector;

a first separator disposed on the first electrode;

a second separator disposed on the second electrode; and

an interlayer disposed between the first separator and the second separator, the interlayer comprising a first layer including a hard carbon material and a second layer different from the first layer, the second layer comprising a cathode material, the interlayer electrically insulated from the first electrode and the second electrode via the first separator and the second separator.

14. The electrochemical cell of claim **1**, wherein at least one of the first separator or the second separator includes a ceramic.

15. The electrochemical cell of claim **2**, wherein the interlayer tab is formed monolithically with the interlayer.

16. The electrochemical cell of claim **1**, wherein the interlayer has a thickness between about 500 nm and about 100 μm.

17. The electrochemical cell of claim **8**, wherein the interlayer includes a binder.

18. The electrochemical cell of claim **8**, wherein the interlayer has a thickness between about 500 nm and about 100 μm.

19. The electrochemical cell of claim **10**, wherein the interlayer includes hard carbon.

20. The electrochemical cell of claim **13**, further comprising:

a first electrode tab extending from the first current collector; and

an interlayer tab extending from the interlayer, the interlayer tab extending beyond the first separator and the second separator.

21. The electrochemical cell of claim **13**, wherein the interlayer is configured to inhibit the projection of dendrites through at least one of the first separator or the second separator during operation of the electrochemical cell.

22. The electrochemical cell of claim **13**, wherein the interlayer is configured to react with any dendritic material that has penetrated at least one of the first separator or the second separator such that the dendritic material dissipates, thus preventing a short circuit.

23. The electrochemical cell of claim **13**, wherein the interlayer has a thickness between about 500 nm and about 100 μm.

24. The electrochemical cell of claim **13**, wherein at least one of the first separator or the second separator includes a ceramic.

* * * * *